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Portable cooler with active temperature control

Abstract

A portable cooler container with active temperature control includes a container body with an insulated chamber to receive medicines, vaccines, biological samples or other medical goods. A control system is at least partially disposed between an outer wall and an inner wall of the container body. An electronic display screen on one of the lid and the container body configured to display shipping address information for the portable cooler container.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1649067	12/1926	Karlson	N/A	N/A
1721311	12/1928	Muenchen	N/A	N/A
1727913	12/1928	Svenn	N/A	N/A
2046125	12/1935	Lacy	N/A	N/A
2483979	12/1948	Morrill	N/A	N/A
2548076	12/1950	Strezoff	N/A	N/A
2746265	12/1955	Mills	N/A	N/A
3064113	12/1961	Pitrone	N/A	N/A
3129116	12/1963	Corry	N/A	N/A
3155260	12/1963	Widener	N/A	N/A
3345934	12/1966	Steiner	N/A	N/A
3435622	12/1968	Barton et al.	N/A	N/A
3463140	12/1968	Rollor, Jr.	N/A	N/A
3536893	12/1969	Cranley	N/A	N/A
3539399	12/1969	Harvey	N/A	N/A
3543842	12/1969	Merges	N/A	N/A
3603106	12/1970	Ryan et al.	N/A	N/A
3607444	12/1970	DeBucs	N/A	N/A
3622753	12/1970	Lax	N/A	N/A
3678248	12/1971	Tricault et al.	N/A	N/A
3739148	12/1972	Ryckman, Jr.	N/A	N/A
3757085	12/1972	Balaguer	N/A	N/A
3766975	12/1972	Todd	N/A	N/A
3797563	12/1973	Hoffmann et al.	N/A	N/A
3823567	12/1973	Corini	N/A	N/A

3892945	12/1974	Lerner	N/A	N/A
3931494	12/1975	Fisher et al.	N/A	N/A
4038831	12/1976	Gaudel et al.	N/A	N/A
4068115	12/1977	Mack	N/A	N/A
4095090	12/1977	Pianezza	N/A	N/A
4134004	12/1978	Anderson et al.	N/A	N/A
4240272	12/1979	Tiede et al.	N/A	N/A
4442343	12/1983	Genuit et al.	N/A	N/A
4470999	12/1983	Carpiac	N/A	N/A
4531046	12/1984	Stover	N/A	N/A
4537044	12/1984	Putnam	N/A	N/A
4681611	12/1986	Bohner	N/A	N/A
4751368	12/1987	Daifotes	N/A	N/A
D296509	12/1987	Fuke	N/A	N/A
4785637	12/1987	Giebeler	N/A	N/A
4801782	12/1988	Ineson	N/A	N/A
4827107	12/1988	Peery	N/A	N/A
4865986	12/1988	Coy et al.	N/A	N/A
4978833	12/1989	Knepler	N/A	N/A
4980539	12/1989	Walton	N/A	N/A
4982722	12/1990	Wyatt	N/A	N/A
4983798	12/1990	Eckler	N/A	N/A
5042258	12/1990	Sundhar	N/A	N/A
5090209	12/1991	Martin	N/A	N/A
5163290	12/1991	Kinnear	N/A	N/A
5199275	12/1992	Martin	N/A	N/A
5208896	12/1992	Katayev	N/A	N/A
5209069	12/1992	Newnan	N/A	N/A
5217064	12/1992	Kellow	N/A	N/A
5243684	12/1992	Edwards	N/A	N/A
5274215	12/1992	Jackson	N/A	N/A
5283420	12/1993	Montalto	N/A	N/A
5313787	12/1993	Martin	N/A	N/A
5343368	12/1993	Miller	N/A	N/A
5388565	12/1994	Ou	N/A	N/A
5448809	12/1994	Kraus	N/A	N/A
5456164	12/1994	Bang	N/A	N/A
5497883	12/1995	Monetti	N/A	N/A
5508494	12/1995	Sarris et al.	N/A	N/A
5508600	12/1995	Myslinski	N/A	N/A
5535815	12/1995	Hyman	N/A	N/A
5549035	12/1995	Wing-Chung	N/A	N/A
5550452	12/1995	Shirai et al.	N/A	N/A
5603220	12/1996	Seaman	N/A	N/A
5603858	12/1996	Wyatt et al.	N/A	N/A
5605047	12/1996	Park	N/A	N/A
5638896	12/1996	Nishino	N/A	N/A
5643485	12/1996	Potter et al.	N/A	N/A
5678925	12/1996	Garmaise et al.	N/A	N/A
5731568	12/1997	Malecek	N/A	N/A

5737923	12/1997	Gilley	N/A	N/A
5771788	12/1997	Lee	N/A	N/A
5786643	12/1997	Wyatt et al.	N/A	N/A
5842353	12/1997	Kuo-Liang	N/A	N/A
5862669	12/1998	Davis	N/A	N/A
5884006	12/1998	Frohlich et al.	N/A	N/A
5903133	12/1998	Amero, Jr. et al.	N/A	N/A
5945651	12/1998	Chorosinski	N/A	N/A
5948301	12/1998	Liebermann	N/A	N/A
5954984	12/1998	Ablah et al.	N/A	N/A
5959433	12/1998	Rohde	N/A	N/A
6000224	12/1998	Foye	N/A	N/A
6000225	12/1998	Ghoshal	N/A	N/A
6003319	12/1998	Gilley et al.	N/A	N/A
6005233	12/1998	Wyatt	N/A	N/A
6013901	12/1999	Lavoie	N/A	N/A
6020575	12/1999	Nagle et al.	N/A	N/A
6032481	12/1999	Mosby	N/A	N/A
6038865	12/1999	Watanabe	N/A	N/A
6042720	12/1999	Reber	N/A	N/A
6072161	12/1999	Stein	N/A	N/A
6075229	12/1999	Vanselow	N/A	N/A
6089409	12/1999	Hart	N/A	N/A
6106784	12/1999	Lund et al.	N/A	N/A
6108489	12/1999	Frohlich et al.	N/A	N/A
6110159	12/1999	Tsujita	N/A	N/A
6119460	12/1999	Huang	N/A	N/A
6123065	12/1999	Teglbjarg	N/A	N/A
6140614	12/1999	Padamsee	N/A	N/A
6141975	12/1999	Tatsumi	N/A	N/A
6144016	12/1999	Garvin	N/A	N/A
6158227	12/1999	Seeley	N/A	N/A
6178753	12/2000	Scudder	N/A	N/A
6180003	12/2000	Reber et al.	N/A	N/A
6209343	12/2000	Owen	N/A	N/A
6212959	12/2000	Perkins	N/A	N/A
6232585	12/2000	Clothier	N/A	N/A
RE37213	12/2000	Staggs	N/A	N/A
6260360	12/2000	Wheeler	N/A	N/A
6274856	12/2000	Clothier	N/A	N/A
6279470	12/2000	Simeray et al.	N/A	N/A
6281611	12/2000	Chen et al.	N/A	N/A
6282906	12/2000	Cauchy	N/A	N/A
6295820	12/2000	Cauchy	N/A	N/A
6308518	12/2000	Hunter	N/A	N/A
6310329	12/2000	Carter	N/A	N/A
6314867	12/2000	Russell	N/A	N/A
6316753	12/2000	Clothier	N/A	N/A
6320169	12/2000	Clothier	N/A	N/A
6350972	12/2001	Wright	N/A	N/A

6351952	12/2001	Baker, III	N/A	N/A
6353208	12/2001	Bostic	N/A	N/A
6376803	12/2001	Klinger	N/A	N/A
6384387	12/2001	Owens	N/A	N/A
6403928	12/2001	Ford	N/A	N/A
6414278	12/2001	Frohlich et al.	N/A	N/A
6415624	12/2001	Connors et al.	N/A	N/A
6427863	12/2001	Nichols	N/A	N/A
6433313	12/2001	Owens	N/A	N/A
6434000	12/2001	Pandolfi	N/A	N/A
6444961	12/2001	Clothier	N/A	N/A
6539725	12/2002	Bell	N/A	N/A
6543335	12/2002	Lassota	N/A	N/A
6555789	12/2002	Owens	N/A	N/A
6558947	12/2002	Lund et al.	N/A	N/A
6571564	12/2002	Upadhye et al.	N/A	N/A
6584374	12/2002	Lee et al.	N/A	N/A
6598405	12/2002	Bell	N/A	N/A
6609392	12/2002	Brown	N/A	N/A
6622515	12/2002	Baker, III	N/A	N/A
6634417	12/2002	Kolowich	N/A	N/A
6637210	12/2002	Bell	N/A	N/A
6651445	12/2002	Clark	N/A	N/A
6657170	12/2002	Clothier	N/A	N/A
6662978	12/2002	Lin et al.	N/A	N/A
6664520	12/2002	Clothier	N/A	N/A
6668577	12/2002	Quenedey	N/A	N/A
6672076	12/2003	Bell	N/A	N/A
6674052	12/2003	Luo	N/A	N/A
6702138	12/2003	Bielecki et al.	N/A	N/A
6703590	12/2003	Holley, Jr.	N/A	N/A
6751963	12/2003	Navedo et al.	N/A	N/A
6753775	12/2003	Auerbach et al.	N/A	N/A
6771183	12/2003	Hunter	N/A	N/A
6818867	12/2003	Kressmann	N/A	N/A
6822198	12/2003	Rix	N/A	N/A
6852954	12/2004	Liu et al.	N/A	N/A
6864462	12/2004	Sanoner et al.	N/A	N/A
6870135	12/2004	Hamm et al.	N/A	N/A
6948321	12/2004	Bell	N/A	N/A
6953913	12/2004	Hara et al.	N/A	N/A
6968888	12/2004	Kolowich	N/A	N/A
7002111	12/2005	Bauer	N/A	N/A
7022946	12/2005	Sanoner et al.	N/A	N/A
7034256	12/2005	Phillips	N/A	N/A
7057527	12/2005	Hunter	N/A	N/A
7059387	12/2005	Kolowich	N/A	N/A
7069739	12/2005	Porter	N/A	N/A
7073678	12/2005	Dibdin et al.	N/A	N/A
7091455	12/2005	Fung	N/A	N/A

7109445	12/2005	Patterson et al.	N/A	N/A
7111465	12/2005	Bell	N/A	N/A
7117684	12/2005	Scudder	N/A	N/A
7140508	12/2005	Kuhn et al.	N/A	N/A
7140768	12/2005	Prabhakar	N/A	N/A
7174720	12/2006	Kennedy	N/A	N/A
7193190	12/2006	Kissel, Jr.	N/A	N/A
7208707	12/2006	Clothier	N/A	N/A
7212955	12/2006	Kirshenbau et al.	N/A	N/A
7225632	12/2006	Derifield	N/A	N/A
7227108	12/2006	Clothier	N/A	N/A
7231771	12/2006	McMurry	N/A	N/A
7260438	12/2006	Caldwell	N/A	N/A
7263283	12/2006	Knepler	N/A	N/A
7263855	12/2006	Meyer et al.	N/A	N/A
7276676	12/2006	Thompson	N/A	N/A
7278270	12/2006	Culp	N/A	N/A
7287386	12/2006	Upadhye et al.	N/A	N/A
7294374	12/2006	Romero	N/A	N/A
7411792	12/2007	Richards et al.	N/A	N/A
7414380	12/2007	Tang et al.	N/A	N/A
7419073	12/2007	Crisp, III	N/A	N/A
7421845	12/2007	Bell	N/A	N/A
7431174	12/2007	Thissen	N/A	N/A
7511617	12/2008	Burman et al.	N/A	N/A
7571830	12/2008	Lin	N/A	N/A
7592084	12/2008	Hoffjann	N/A	N/A
7659493	12/2009	Reusche et al.	N/A	N/A
7681754	12/2009	Ross	N/A	N/A
7683572	12/2009	Toya	N/A	N/A
7721566	12/2009	Wilken	N/A	N/A
7728711	12/2009	Shoenfeld	N/A	N/A
7748223	12/2009	Minoura	N/A	N/A
7764497	12/2009	Becklin	N/A	N/A
7784301	12/2009	Sasaki et al.	N/A	N/A
7802446	12/2009	Overgaard	N/A	N/A
7815067	12/2009	Matsumoto et al.	N/A	N/A
7825353	12/2009	Shingler	N/A	N/A
7836722	12/2009	Magill et al.	N/A	N/A
7861538	12/2010	Welle et al.	N/A	N/A
7872214	12/2010	Schandel	N/A	N/A
7886655	12/2010	Lassota	N/A	N/A
7908870	12/2010	Williams et al.	N/A	N/A
7913511	12/2010	Meyer et al.	N/A	N/A
7926293	12/2010	Bell	N/A	N/A
7934537	12/2010	Kolowich	N/A	N/A
7939312	12/2010	Roberts et al.	N/A	N/A
7942145	12/2010	Palena et al.	N/A	N/A
7948209	12/2010	Jung	N/A	N/A
7966927	12/2010	Yoakim	N/A	N/A

7997786	12/2010	Liu	N/A	N/A
8055310	12/2010	Beart et al.	N/A	N/A
8056357	12/2010	Bruce	N/A	N/A
8061149	12/2010	Gowans	N/A	N/A
8076620	12/2010	Maupin et al.	N/A	N/A
8113365	12/2011	Brown	N/A	N/A
8146485	12/2011	Ozanne	N/A	N/A
8156755	12/2011	Murray	N/A	N/A
8205468	12/2011	Hemminger et al.	N/A	N/A
8215835	12/2011	Hyde et al.	N/A	N/A
8272530	12/2011	Rebernik	N/A	N/A
8272532	12/2011	Michaelian et al.	N/A	N/A
8274016	12/2011	Montana	N/A	N/A
8280453	12/2011	Beart et al.	N/A	N/A
8319154	12/2011	Shaikh et al.	N/A	N/A
8336729	12/2011	Kelly	N/A	N/A
8362351	12/2012	Hagg et al.	N/A	N/A
8375728	12/2012	Bell	N/A	N/A
8391104	12/2012	De la Huerga	N/A	N/A
8398602	12/2012	Ilo	N/A	N/A
8400104	12/2012	Adamczyk et al.	N/A	N/A
8424316	12/2012	Tuszkiewicz	N/A	N/A
8448457	12/2012	Cutting et al.	N/A	N/A
8448809	12/2012	Kelly	N/A	N/A
8453477	12/2012	Crespo et al.	N/A	N/A
8467669	12/2012	Widanagamage et al.	N/A	N/A
8479941	12/2012	Matsumoto et al.	N/A	N/A
8539790	12/2012	Budd	N/A	N/A
8607581	12/2012	Williams et al.	N/A	N/A
8618448	12/2012	Alexander	N/A	N/A
8621980	12/2013	Bunn	N/A	N/A
8646282	12/2013	Llercil	N/A	N/A
8659903	12/2013	Schwartz	N/A	N/A
8677767	12/2013	Ilercil et al.	N/A	N/A
8759721	12/2013	Alexander	N/A	N/A
D715143	12/2013	Hewitt	N/A	N/A
8887512	12/2013	Olsen	N/A	N/A
8887944	12/2013	Deane et al.	N/A	N/A
8893513	12/2013	June	N/A	N/A
8904809	12/2013	Yuan et al.	N/A	N/A
8907796	12/2013	Sweeney et al.	N/A	N/A
8919138	12/2013	Kobayashi	N/A	N/A
8938986	12/2014	Matta et al.	N/A	N/A
8991194	12/2014	Edwards et al.	N/A	N/A
9021825	12/2014	Hewitt	N/A	N/A
9022249	12/2014	Ranade	N/A	N/A
9035222	12/2014	Alexander	N/A	N/A
9057568	12/2014	Malik et al.	N/A	N/A
9060508	12/2014	Anti et al.	N/A	N/A

9103572	12/2014	Edwards et al.	N/A	N/A
9115919	12/2014	Ilercil	N/A	N/A
9134055	12/2014	Ilercil	N/A	N/A
9138295	12/2014	Hyde et al.	N/A	N/A
9139319	12/2014	Crespo et al.	N/A	N/A
9139351	12/2014	Chou et al.	N/A	N/A
9140476	12/2014	Eckhoff et al.	N/A	N/A
9144180	12/2014	Olsson et al.	N/A	N/A
9151523	12/2014	Ilercil	N/A	N/A
9151545	12/2014	Soukhojak	N/A	N/A
9182155	12/2014	Crumlin	N/A	N/A
9184427	12/2014	Chuang	N/A	N/A
9272475	12/2015	Ranade et al.	N/A	N/A
9310111	12/2015	Edwards et al.	N/A	N/A
9341394	12/2015	Edwards et al.	N/A	N/A
9351600	12/2015	Rime	N/A	N/A
9366469	12/2015	Chapman, Jr.	N/A	N/A
9372016	12/2015	Bloedow et al.	N/A	N/A
9424548	12/2015	Siegel	N/A	N/A
9429350	12/2015	Chapman, Jr.	N/A	N/A
9435578	12/2015	Calderon et al.	N/A	N/A
9447995	12/2015	Bloedow et al.	N/A	N/A
9470440	12/2015	Ilercil	N/A	N/A
9480363	12/2015	Delattre	N/A	N/A
9513067	12/2015	Ahmed	N/A	N/A
9573754	12/2016	Ahmed et al.	N/A	N/A
9581362	12/2016	Stanley et al.	N/A	N/A
9593871	12/2016	Stanley et al.	N/A	N/A
9599376	12/2016	Ilercil	N/A	N/A
9618253	12/2016	Tansley	N/A	N/A
9685598	12/2016	Marc	N/A	N/A
9688454	12/2016	Ranade	N/A	N/A
9713798	12/2016	Hewitt	N/A	N/A
9752808	12/2016	Nakamura	N/A	N/A
9758299	12/2016	Ahmed et al.	N/A	N/A
9791184	12/2016	Novisoff et al.	N/A	N/A
9791185	12/2016	Ilercil	N/A	N/A
9795979	12/2016	Adler	N/A	N/A
9802806	12/2016	Hewitt	N/A	N/A
9814331	12/2016	Alexander	N/A	N/A
9828165	12/2016	Ranade et al.	N/A	N/A
9829221	12/2016	Ilercil	N/A	N/A
9874377	12/2017	Ilercil	N/A	N/A
9885502	12/2017	Yuan et al.	N/A	N/A
9950851	12/2017	Ranade	N/A	N/A
9958187	12/2017	Monroy	N/A	N/A
10012417	12/2017	Edwards et al.	N/A	N/A
10101420	12/2017	Wikus et al.	N/A	N/A
10119733	12/2017	Ilercil	N/A	N/A
10131478	12/2017	Maser	N/A	N/A

10156388	12/2017	Ilercil	N/A	N/A
10161657	12/2017	Ilercil	N/A	N/A
10181109	12/2018	Joao	N/A	N/A
10188229	12/2018	Alexander	N/A	N/A
10274241	12/2018	Ghiraldi	N/A	N/A
10279979	12/2018	Ranade	N/A	N/A
10287085	12/2018	Kuhn	N/A	N/A
10328074	12/2018	Engelhardt et al.	N/A	N/A
10372922	12/2018	Paterra	N/A	N/A
10405650	12/2018	Turner et al.	N/A	N/A
10458684	12/2018	Ilercil	N/A	N/A
10472158	12/2018	Ranade	N/A	N/A
10495357	12/2018	Ilercil	N/A	N/A
10549900	12/2019	McCormick	N/A	N/A
10562695	12/2019	Knight et al.	N/A	N/A
10625922	12/2019	Epenetos et al.	N/A	N/A
10722427	12/2019	Cantor	N/A	A61J 1/165
10743708	12/2019	Alexander et al.	N/A	N/A
10823478	12/2019	Williams	N/A	N/A
10850047	12/2019	McCormick	N/A	N/A
11083332	12/2020	Alexander et al.	N/A	N/A
11089891	12/2020	Alexander	N/A	N/A
11090225	12/2020	Vlahinos et al.	N/A	N/A
2001/0009609	12/2000	Bradenbaugh	N/A	N/A
2001/0022304	12/2000	Roche	N/A	N/A
2001/0023866	12/2000	Wang	N/A	N/A
2002/0001297	12/2001	Yamagiwa	N/A	N/A
2002/0023912	12/2001	Mcgee	N/A	N/A
2002/0083840	12/2001	Lassota	N/A	N/A
2002/0104318	12/2001	Jaafar	N/A	N/A
2002/0129712	12/2001	Westbrook	N/A	N/A
2002/0162339	12/2001	Harrison	N/A	N/A
2002/0175158	12/2001	Sanoner et al.	N/A	N/A
2003/0010768	12/2002	Li	N/A	N/A
2003/0024250	12/2002	Haas	N/A	N/A
2003/0029862	12/2002	Clothier	N/A	N/A
2003/0029876	12/2002	Giraud	N/A	N/A
2003/0066638	12/2002	Qu	N/A	N/A
2003/0074903	12/2002	Upadhye	N/A	N/A
2003/0122455	12/2002	Caldwell	N/A	N/A
2003/0145621	12/2002	Kidwell	N/A	N/A
2004/0004072	12/2003	Clothier	N/A	N/A
2004/0006996	12/2003	Butcher	N/A	N/A
2004/0007553	12/2003	Smolko	N/A	N/A
2004/0111179	12/2003	Broadfield et al.	N/A	N/A
2004/0159240	12/2003	Lyall, III	N/A	N/A
2004/0167592	12/2003	Grove	N/A	N/A
2004/0194470	12/2003	Upadhye et al.	N/A	N/A
2004/0212120	12/2003	Giraud	N/A	N/A

2004/0226309	12/2003	Broussard	N/A	N/A
2005/0045615	12/2004	Sanoner et al.	N/A	N/A
2005/0045618	12/2004	Ito	N/A	N/A
2005/0121431	12/2004	Yuen	N/A	N/A
2005/0242804	12/2004	Hintz	N/A	N/A
2006/0005873	12/2005	Kambe et al.	N/A	N/A
2006/0021513	12/2005	Ide	N/A	N/A
2006/0023480	12/2005	Plummer	N/A	N/A
2006/0081599	12/2005	Anderson	N/A	N/A
2006/0173259	12/2005	Flaherty	N/A	N/A
2006/0207442	12/2005	Pettersson	N/A	N/A
2006/0209628	12/2005	Jones	N/A	N/A
2006/0261233	12/2005	Williams et al.	N/A	N/A
2007/0024237	12/2006	Cole et al.	N/A	N/A
2007/0051727	12/2006	Holley	N/A	N/A
2007/0092773	12/2006	Guo	N/A	N/A
2007/0144205	12/2006	Moore	N/A	N/A
2007/0151457	12/2006	Rabin et al.	N/A	N/A
2007/0182367	12/2006	Partovi	N/A	N/A
2007/0186577	12/2006	Goncharko	N/A	N/A
2007/0193297	12/2006	Wilson	N/A	N/A
2007/0223895	12/2006	Flemm	N/A	N/A
2007/0257766	12/2006	Richards et al.	N/A	N/A
2007/0278207	12/2006	Van Hoy	N/A	N/A
2007/0279002	12/2006	Partovi	N/A	N/A
2008/0011077	12/2007	Ramus et al.	N/A	N/A
2008/0019122	12/2007	Kramer	N/A	N/A
2008/0022695	12/2007	Welle	N/A	N/A
2008/0022696	12/2007	Welle	N/A	N/A
2008/0041233	12/2007	Bunn	N/A	N/A
2008/0041859	12/2007	Teglbjarg	N/A	N/A
2008/0087270	12/2007	Shaikh	N/A	N/A
2008/0121630	12/2007	Simard	N/A	N/A
2008/0135564	12/2007	Romero	N/A	N/A
2008/0141681	12/2007	Arnold	N/A	N/A
2008/0149624	12/2007	Tamura	N/A	N/A
2008/0179311	12/2007	Koro et al.	N/A	N/A
2008/0190914	12/2007	Gibson	N/A	N/A
2008/0213449	12/2007	Wisner et al.	N/A	N/A
2008/0251063	12/2007	Palena et al.	N/A	N/A
2008/0272134	12/2007	Rohe	N/A	N/A
2009/0049845	12/2008	McStravick et al.	N/A	N/A
2009/0058352	12/2008	Lin	N/A	N/A
2009/0064687	12/2008	Tuszkiewicz	N/A	N/A
2009/0071952	12/2008	Kuwabara	N/A	N/A
2009/0078708	12/2008	Williams	N/A	N/A
2009/0102296	12/2008	Greene et al.	N/A	N/A
2009/0152276	12/2008	Groll	N/A	N/A
2009/0158770	12/2008	Cohrs et al.	N/A	N/A
2009/0166350	12/2008	Ho	N/A	N/A

2009/0184102	12/2008	Parker, Jr. et al.	N/A	N/A
2009/0200320	12/2008	Saito	N/A	N/A
2009/0230117	12/2008	Fernando	N/A	N/A
2009/0277187	12/2008	McGann	N/A	N/A
2010/0000980	12/2009	Popescu	N/A	N/A
2010/0028758	12/2009	Eaves	N/A	N/A
2010/0064698	12/2009	Schabron	N/A	N/A
2010/0089247	12/2009	Yang	N/A	N/A
2010/0108694	12/2009	Sedlbauer et al.	N/A	N/A
2010/0125417	12/2009	Hyde et al.	N/A	N/A
2010/0145688	12/2009	Sung et al.	N/A	N/A
2010/0147014	12/2009	Kim	N/A	N/A
2010/0158489	12/2009	Siu et al.	N/A	N/A
2010/0158660	12/2009	Radhakrishnan	N/A	N/A
2010/0186499	12/2009	Ramus et al.	N/A	N/A
2010/0251755	12/2009	Lauchnor	N/A	N/A
2010/0299278	12/2009	Kriss	N/A	N/A
2011/0041546	12/2010	Linder	N/A	N/A
2011/0056215	12/2010	Ham et al.	N/A	N/A
2011/0062149	12/2010	Oriel	N/A	N/A
2011/0070474	12/2010	Lee et al.	N/A	N/A
2011/0072978	12/2010	Popescu	N/A	N/A
2011/0073603	12/2010	Saho et al.	N/A	N/A
2011/0082437	12/2010	Stacey	N/A	N/A
2011/0108506	12/2010	Lindhorst-Ko	N/A	N/A
2011/0121660	12/2010	Azancot et al.	N/A	N/A
2011/0143000	12/2010	Fiset	N/A	N/A
2011/0152979	12/2010	Driscoll et al.	N/A	N/A
2011/0155621	12/2010	Lindquist et al.	N/A	N/A
2011/0174993	12/2010	Blain	N/A	N/A
2011/0179807	12/2010	Holloway	N/A	N/A
2011/0180527	12/2010	Abbott	N/A	N/A
2011/0198255	12/2010	Baumfalk et al.	N/A	N/A
2011/0247356	12/2010	Krosse et al.	N/A	N/A
2011/0259871	12/2010	Li	N/A	N/A
2011/0265562	12/2010	Li	N/A	N/A
2012/0061050	12/2011	Petrillo et al.	N/A	N/A
2012/0064470	12/2011	Delattre et al.	N/A	N/A
2012/0082766	12/2011	Maupin et al.	N/A	N/A
2012/0090333	12/2011	DellaMorte et al.	N/A	N/A
2012/0103562	12/2011	Alexander	N/A	N/A
2012/0118874	12/2011	Williams et al.	N/A	N/A
2012/0132646	12/2011	England et al.	N/A	N/A
2012/0138597	12/2011	Quella et al.	N/A	N/A
2012/0152511	12/2011	Chang et al.	N/A	N/A
2012/0193999	12/2011	Zeine	N/A	N/A
2012/0235505	12/2011	Schatz et al.	N/A	N/A
2012/0235636	12/2011	Partovi	N/A	N/A
2012/0248095	12/2011	Lee et al.	N/A	N/A
2012/0248096	12/2011	Lee et al.	N/A	N/A

2012/0255946	12/2011	Kim et al.	N/A	N/A
2012/0256585	12/2011	Partovi et al.	N/A	N/A
2012/0258229	12/2011	Mindrup	N/A	N/A
2012/0285180	12/2011	Wilkinson	N/A	N/A
2012/0312031	12/2011	Olsen	N/A	N/A
2012/0319500	12/2011	Beart et al.	N/A	N/A
2013/0059259	12/2012	Oldani	N/A	N/A
2013/0103463	12/2012	Briar et al.	N/A	N/A
2013/0128915	12/2012	Aschauer et al.	N/A	N/A
2013/0167730	12/2012	Behm	N/A	N/A
2013/0180563	12/2012	Makansi	N/A	N/A
2013/0200064	12/2012	Alexander	N/A	N/A
2013/0206015	12/2012	Jacoby et al.	N/A	N/A
2013/0221013	12/2012	Kolowich et al.	N/A	N/A
2013/0239607	12/2012	Kelly	N/A	N/A
2013/0245991	12/2012	Kriss	N/A	N/A
2013/0255306	12/2012	Mayer	N/A	N/A
2013/0255824	12/2012	Williams et al.	N/A	N/A
2013/0273752	12/2012	Rudisill	N/A	N/A
2013/0275075	12/2012	Johnson	N/A	N/A
2013/0287967	12/2012	Ono	N/A	N/A
2013/0306656	12/2012	Eckhoff	N/A	N/A
2014/0137570	12/2013	Hauck et al.	N/A	N/A
2014/0150464	12/2013	Bloedow	N/A	N/A
2014/0165607	12/2013	Alexander	N/A	N/A
2014/0230484	12/2013	Yavitz	N/A	N/A
2014/0238985	12/2013	Sweeney et al.	N/A	N/A
2014/0298492	12/2013	Simpson	N/A	N/A
2014/0305927	12/2013	Alexander	N/A	N/A
2014/0318153	12/2013	Ilercil	N/A	N/A
2014/0338713	12/2013	Nakanuma	N/A	N/A
2014/0340197	12/2013	Flood	N/A	N/A
2014/0352329	12/2013	Bloedow et al.	N/A	N/A
2015/0024349	12/2014	Bischoff	N/A	N/A
2015/0122688	12/2014	Dias	N/A	N/A
2015/0143823	12/2014	Slack et al.	N/A	N/A
2015/0161871	12/2014	Kim	N/A	N/A
2015/0205625	12/2014	Pearson et al.	N/A	N/A
2015/0245723	12/2014	Alexander	N/A	N/A
2015/0321195	12/2014	Malik et al.	N/A	N/A
2015/0335184	12/2014	Balachandran	N/A	N/A
2015/0349233	12/2014	Span et al.	N/A	N/A
2016/0035957	12/2015	Casey	N/A	N/A
2016/0111622	12/2015	Lee et al.	N/A	N/A
2016/0183730	12/2015	Bedi	N/A	N/A
2016/0271015	12/2015	Wengreen et al.	N/A	N/A
2017/0042373	12/2016	Alexander et al.	N/A	N/A
2017/0056289	12/2016	Tsuno	N/A	N/A
2017/0108261	12/2016	Broussard	N/A	N/A
2017/0150840	12/2016	Park	N/A	N/A

2017/0177883	12/2016	Paterra et al.	N/A	N/A
2017/0178069	12/2016	Paterra	N/A	N/A
2017/0180368	12/2016	Paterra	N/A	N/A
2017/0206497	12/2016	Kriss	N/A	N/A
2017/0259956	12/2016	Hori et al.	N/A	N/A
2017/0271570	12/2016	Marc	N/A	N/A
2017/0290741	12/2016	Chou et al.	N/A	N/A
2017/0314851	12/2016	Alexander et al.	N/A	N/A
2017/0336134	12/2016	Williams	N/A	N/A
2017/0372260	12/2016	Desmarais et al.	N/A	N/A
2018/0023865	12/2017	Llercil	N/A	N/A
2018/0036202	12/2017	Wengreen et al.	N/A	N/A
2018/0039940	12/2017	Varga	N/A	N/A
2018/0061162	12/2017	High et al.	N/A	N/A
2018/0075753	12/2017	Joao	N/A	N/A
2018/0164034	12/2017	Banks	N/A	N/A
2018/0175272	12/2017	Imai et al.	N/A	N/A
2018/0225464	12/2017	Paterra	N/A	N/A
2018/0320947	12/2017	Jain et al.	N/A	N/A
2018/0325296	12/2017	Lavi	N/A	N/A
2018/0327165	12/2017	Lee, Sr. et al.	N/A	N/A
2018/0352796	12/2017	Chattman	N/A	N/A
2018/0353379	12/2017	Chou et al.	N/A	N/A
2018/0360264	12/2017	Alexander	N/A	N/A
2019/0003757	12/2018	Miros et al.	N/A	N/A
2019/0003781	12/2018	Caniere et al.	N/A	N/A
2019/0039811	12/2018	Kuhn et al.	N/A	N/A
2019/0112119	12/2018	Alexander	N/A	N/A
2019/0263219	12/2018	Spath	N/A	N/A
2019/0277553	12/2018	Vlahinos	N/A	N/A
2019/0303862	12/2018	Bollinger et al.	N/A	N/A
2019/0359411	12/2018	Fallgren	N/A	N/A
2019/0373755	12/2018	Paterra	N/A	N/A
2019/0390890	12/2018	Alexander et al.	N/A	N/A
2020/0041197	12/2019	Kim	N/A	N/A
2020/0045959	12/2019	Hoffmann et al.	N/A	N/A
2020/0224964	12/2019	Alexander	N/A	N/A
2020/0309442	12/2019	Scully, Jr. et al.	N/A	N/A
2020/0361690	12/2019	Sanders et al.	N/A	N/A
2021/0169740	12/2020	Janzen et al.	N/A	N/A
2023/0025634	12/2022	Kim et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
631614	12/1981	CH	N/A
1338240	12/2001	CN	N/A
1502513	12/2003	CN	N/A
2708795	12/2004	CN	N/A
1748112	12/2005	CN	N/A

1776992	12/2005	CN	N/A
2922666	12/2006	CN	N/A
101069606	12/2006	CN	N/A
101109795	12/2007	CN	N/A
201042350	12/2007	CN	N/A
201076180	12/2007	CN	N/A
201308643	12/2007	CN	N/A
201237271	12/2008	CN	N/A
101507261	12/2008	CN	N/A
201303850	12/2008	CN	N/A
201445353	12/2009	CN	N/A
101820128	12/2009	CN	N/A
201612420	12/2009	CN	N/A
102164 526	12/2010	CN	N/A
102802294	12/2011	CN	N/A
202681700	12/2012	CN	N/A
202919767	12/2012	CN	N/A
102266184	12/2012	CN	N/A
203468187	12/2013	CN	N/A
106102460	12/2015	CN	N/A
108974637	12/2017	CN	N/A
19744526	12/1998	DE	N/A
20108363	12/2000	DE	N/A
20314416	12/2003	DE	N/A
0332355	12/1988	EP	N/A
0722708	12/1995	EP	N/A
0895772	12/1998	EP	N/A
1034407	12/1999	EP	N/A
1701313	12/2005	EP	N/A
2022727	12/2008	EP	N/A
2165243	12/2009	EP	N/A
2273250	12/2010	EP	N/A
2001761	12/2011	EP	N/A
2308771	12/2011	EP	N/A
2852540	12/2015	EP	N/A
3109574	12/2015	EP	N/A
3295098	12/2017	EP	N/A
3189944	12/2017	EP	N/A
3531354	12/2018	EP	N/A
3491301	12/2019	EP	N/A
2737380	12/1996	FR	N/A
2752377	12/1997	FR	N/A
2763463	12/1997	FR	N/A
2828082	12/2002	FR	N/A
1311955	12/1972	GB	N/A
2304179	12/1996	GB	N/A
2390798	12/2003	GB	N/A
2414922	12/2004	GB	N/A
2441825	12/2007	GB	N/A
S54-147575	12/1952	JP	N/A

S63-249519	12/1987	JP	N/A
H01-164322	12/1988	JP	N/A
H04-201877	12/1991	JP	N/A
H5-118725	12/1992	JP	N/A
H05-306472	12/1992	JP	N/A
H06-021549	12/1993	JP	N/A
H1-076418	12/1997	JP	N/A
H10-146276	12/1997	JP	N/A
11-268777	12/1998	JP	N/A
2000-279302	12/1999	JP	N/A
2001524657	12/2000	JP	N/A
2002-516228	12/2001	JP	N/A
3302626	12/2001	JP	N/A
2003-106728	12/2002	JP	N/A
2003-299255	12/2002	JP	N/A
3102537	12/2003	JP	N/A
2004-261493	12/2003	JP	N/A
3594343	12/2003	JP	N/A
H8-142543	12/2003	JP	N/A
2006-345957	12/2004	JP	N/A
2005-308353	12/2004	JP	N/A
2006-068152	12/2005	JP	N/A
2006-102234	12/2005	JP	N/A
2006-166522	12/2005	JP	N/A
2007-064557	12/2006	JP	N/A
2007-139328	12/2006	JP	N/A
2007-260838	12/2006	JP	N/A
2007-312932	12/2006	JP	N/A
2008-173464	12/2007	JP	N/A
3153007	12/2008	JP	N/A
2009288234	12/2008	JP	N/A
2010-527226	12/2009	JP	N/A
2011-171205	12/2010	JP	N/A
2012-523085	12/2011	JP	N/A
5127819	12/2012	JP	N/A
5481388	12/2013	JP	N/A
2017122521	12/2016	JP	N/A
2018-012510	12/2017	JP	N/A
2018-512526	12/2017	JP	N/A
2019-516944	12/2018	JP	N/A
200390049	12/2004	KR	N/A
20100124932	12/2009	KR	N/A
10-2015-0051074	12/2014	KR	N/A
101738787	12/2016	KR	N/A
95021361	12/1994	WO	N/A
WO 02/067737	12/2001	WO	N/A
WO 2003/073030	12/2002	WO	N/A
WO 2004/055654	12/2003	WO	N/A
WO 2008/028329	12/2007	WO	N/A
WO 2008/065175	12/2007	WO	N/A

WO 2008/137996	12/2007	WO	N/A
WO 2008/155538	12/2007	WO	N/A
WO 2009/138930	12/2008	WO	N/A
WO 2010/087560	12/2009	WO	N/A
WO 2010/087560	12/2009	WO	N/A
WO 2010/132627	12/2009	WO	N/A
WO 2011/131595	12/2010	WO	N/A
WO 2012/075449	12/2011	WO	N/A
WO 2012/104665	12/2011	WO	N/A
WO 2013/187763	12/2012	WO	N/A
WO 2014/158655	12/2013	WO	N/A
2015125790	12/2014	WO	N/A
WO 2016/166324	12/2015	WO	N/A
WO 2016/168523	12/2015	WO	N/A
WO 2016/193480	12/2015	WO	N/A
WO 2018/016238	12/2017	WO	N/A
WO 2018/197049	12/2017	WO	N/A
WO 2018/222645	12/2017	WO	N/A
WO 2019/204660	12/2018	WO	N/A

OTHER PUBLICATIONS

Australian Examination Report regarding Application No. 2016216669, dated Feb. 14, 2019, four pages. cited by applicant

Chinese Office Action, regarding Application No. 201510869257.5, dated Aug. 30, 2018, 9 pages. cited by applicant

Chinese Office Action dated Jan. 19, 2023, received in Chinese Patent Application No. 201980033275.6, in 9 pages. cited by applicant

Decision of Rejection mailed on Apr. 4, 2017 in JP Application No. 2013-537797. cited by applicant

European Office Action dated Sep. 28, 2017, received in European Patent Application No. 14 774 350.4, pp. 5. cited by applicant

European Patent Office Communication, regarding Application No. 19 721 954.6, dated Apr. 11, 2022, 7 pages. cited by applicant

European Patent Office Search Report dated Mar. 17, 2016 regarding Application No. 11838764.6-1804, PCT/US2011059014, 7 pages. cited by applicant

European Search Report received in European Patent Application No. 15811173.2, dated Dec. 13, 2017. cited by applicant

First Office Action mailed on Nov. 23, 2016 in CN Application No. 201480014620.9. cited by applicant

International Preliminary Report on Patentability mailed on May 7, 2013 in PCT Application No. PCT/US2011/059014. cited by applicant

International Search and Written Opinion dated Jul. 9, 2019, received in International Patent Application No. PCT/US2019/028198. cited by applicant

International Search Report and Written Opinion dated Aug. 12, 2020, received in International Patent Application No. PCT/US2020/012591, 9 pages. cited by applicant

International Search Report and Written Opinion mailed on Jan. 12, 2016 in PCT Application No. PCT/US15/36304. cited by applicant

International Search Report and Written Opinion mailed on Dec. 9, 2014 in PCT/US2014/019130. cited by applicant

International Search Report and Written Opinion mailed on Jul. 12, 2017, in PCT Application No.

PCT/US2017/031534. cited by applicant
International Search Report and Written Opinion mailed on Mar. 16, 2012 in PCT/US2011/059014. cited by applicant
Japanese Office Action dated Dec. 16, 2022, received in Japanese Patent Application No. 2020-557261, in 18 pages. cited by applicant
Non-final Office Action mailed on Nov. 14, 2016 in U.S. Appl. No. 15/050,714. cited by applicant
Non-final office action mailed on Aug. 2, 2016 in Japanese Patent Application No. 2013-537797. cited by applicant
Notice of Reason(s) for Rejection mailed on Aug. 11, 2015 in JP Application No. 2013-537797. cited by applicant
Office Action dated Aug. 7, 2018, received for Japanese Patent Application No. JP 2017-151497, 4 pages. cited by applicant
Office Action dated Jan. 12, 2018, received in Chinese Application No. 201510869257.5. cited by applicant
Office Action in related Chinese Application No. 201180063844.5, mailed Dec. 29, 2014. cited by applicant
Office Action mailed Sep. 4, 2018, regarding Japan Patent Application No. 2017-554610, 10 pages. cited by applicant
Office Action received in Japanese Patent Application No. 2017-151497, dated Nov. 21, 2017, 5 pages. cited by applicant
Patent Examination Report No. 1 in related Australian Application No. 2011323416, mailed May 15, 2015. cited by applicant
Patent Examination Report No. 2 in related Australian Application No. 2011323416, mailed on Oct. 20, 2015. cited by applicant
PCT International Search Report and Written Opinion dated Sep. 14, 2017 regarding International Application No. PCT/US2017/034081, 15 pages. cited by applicant
PCT International Search Report and Written Opinion mailed on Aug. 17, 2017 in PCT Application No. PCT/US2017/032020. cited by applicant
Second Office Action mailed on Apr. 10, 2017 in CN Application No. 201510869257.5. cited by applicant
Supplementary European Search Report mailed on Oct. 18, 2016 in European Patent Application No. 14 77 4350. cited by applicant
Nov. 25, 2024—(AU) Examination Report No. 2—App. No. 2019256534. cited by applicant
Jan. 13, 2025—(AU) Examination Report No. 3—App. No. 2019256534. cited by applicant
Jan. 15, 2024—(AU) First Examination Report—App. No. 2019256534. cited by applicant
Dec. 3, 2024—(JP) First Office Action—App. No. 2024-014490. cited by applicant
Jan. 7, 2025—(US) Final Office Action—U.S. Appl. No. 18/774,779. cited by applicant

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Background/Summary

INCORPORATION BY REFERENCE TO ANY PRIORITY APPLICATIONS

(1) Any and all applications for which a foreign or domestic priority claim is identified in the Application Data Sheet as filed with the present application are hereby incorporated by reference under 37 CFR 1.57 and should be considered a part of this specification.

BACKGROUND OF THE INVENTION

Field of the Invention

(2) The invention is directed to a portable cooler (e.g., for medicine such as insulin, vaccines, epinephrine, medicine injectors, cartridges, biological fluids, etc.), and more particularly to a portable cooler with active temperature control.

Description of the Related Art

(3) Certain medicine needs to be maintained at a certain temperature or temperature range to be effective (e.g., to maintain potency). Once potency of medicine (e.g., a vaccine) is lost, it cannot be restored, rendering the medicine ineffective and/or unusable. However, maintaining the cold chain (e.g., a record of the medicine's temperature history as it travels through various distribution channels) can be difficult. Additionally, where medicine is transported to remote locations for delivery (e.g., rural, mountainous, sparsely populated areas without road access), maintaining the medicine in the required temperature range may be difficult, especially when travelling through harsh (e.g., desert) climates. Existing medicine transport coolers are passive and inadequate for proper cold chain control (e.g., when used in extreme weather, such as in desert climates, tropical or subtropical climates, etc.).

SUMMARY

(4) Accordingly, there is a need for improved portable cooler designs (e.g., for transporting medicine, such as vaccines, insulin, epinephrine, vials, cartridges, injector pens, etc.) that can maintain the contents of the cooler at a desired temperature or temperature range. Additionally, there is a need for an improved portable cooler design with improved cold chain control and record keeping of the temperature history of the contents (e.g., medicine, such as vaccines) of the cooler (e.g., during transport to remote locations).

(5) In accordance with one aspect, a portable cooler container with active temperature control system is provided. The active temperature control system is operated to heat or cool a chamber of a vessel to approach a temperature set point suitable for a medication stored in the cooler container.

(6) In accordance with another aspect, a portable cooler is provided that includes a temperature control system operable (e.g., automatically) to maintain the chamber of the cooler at a desired temperature or temperature range for a prolonged period of time. Optionally, the portable cooler is sized to house one or more liquid containers (e.g., medicine vials, cartridges or containers, such as a vaccine vials or insulin vials/cartridges, medicine injectors). Optionally, the portable cooler automatically logs (e.g., stores on a memory of the cooler) and/or communicates data on one or more sensed parameters (e.g., of the temperature of the chamber) to a remote electronic device (e.g., remote computer, mobile electronic device such as a smartphone or tablet computer, remote server, etc.). Optionally, the portable cooler can automatically log and/or transmit the data to the remote electronic device (e.g., automatically in real time, periodically at set intervals, etc.).

(7) In accordance with another aspect, a portable cooler container with active temperature control is provided. The container comprises a container body having a chamber configured to receive and hold one or more volumes of perishable liquid, the chamber defined by a base and an inner peripheral wall of the container body. The container also comprises a temperature control system comprising one or more thermoelectric elements configured to actively heat or cool at least a portion of the chamber, and circuitry configured to control an operation of the one or more thermoelectric elements to heat or cool at least a portion of the chamber to a predetermined temperature or temperature range.

(8) Optionally, the container can include one or more batteries configured to provide power to one or both of the circuitry and the one or more thermoelectric elements.

(9) Optionally, the circuitry is further configured to wirelessly communicate with a cloud-based data storage system and/or a remote electronic device.

(10) Optionally, the container includes a first heat sink in communication with the chamber, the first sink being selectively thermally coupled to the one or more thermoelectric elements.

- (11) Optionally, the container includes a second heat sink in communication with the one or more thermoelectric elements (TECs), such that the one or more TECs are disposed between the first heat sink and the second heat sink.
- (12) Optionally, the second heat sink is in thermal communication with a fan operable to draw heat from the second heat sink.
- (13) In one implementation, such as where the ambient temperature is above the predetermined temperature or temperature range, the temperature control system is operable to draw heat from the chamber via the first heat sink, which transfers said heat to the one or more TECs, which transfer said heat to the second heat sink, where the optional fan dissipates heat from the second heat sink.
- (14) In another implementation, such as where the ambient temperature is below the predetermined temperature or temperature range, the temperature control system is operable to add heat to the chamber via the first heat sink, which transfers said heat from the one or more TECs.
- (15) In accordance with one aspect of the disclosure, a portable cooler container with active temperature control is provided. The portable cooler container comprises a container body having a chamber configured to receive and hold one or more containers (e.g., of medicine). The portable cooler container also comprises a lid removably coupleable to the container body to access the chamber, and a temperature control system. The temperature control system comprises one or more thermoelectric elements configured to actively heat or cool at least a portion of the chamber, one or more batteries and circuitry configured to control an operation of the one or more thermoelectric elements to heat or cool at least a portion of the chamber to a predetermined temperature or temperature range. A display screen is disposed on one or both of the container body and the lid, the display screen configured to selectively display shipping information for the portable cooler container using electronic ink.
- (16) In accordance with another aspect of the disclosure, a portable cooler container with active temperature control is provided. The portable cooler container comprises a container body having a chamber configured to receive and hold one or more containers (e.g., of medicine), the chamber defined by a base and an inner peripheral wall of the container body. A lid is removably coupleable to the container body to access the chamber. The portable cooler container also comprises a temperature control system. The temperature control system comprises one or more thermoelectric elements and one or more fans, one or both of the thermoelectric elements and fans configured to actively heat or cool at least a portion of the chamber, one or more batteries and circuitry configured to control an operation of the one or more thermoelectric elements to heat or cool at least a portion of the chamber to a predetermined temperature or temperature range.
- (17) In accordance with another aspect of the disclosure, a portable cooler container with active temperature control is provided. The portable cooler container comprises a container body having a chamber configured to receive and hold one or more volumes of perishable liquid, the chamber defined by a base and an inner peripheral wall of the container body, and a lid movably coupled to the container body by one or more hinges. The portable cooler container also comprises a temperature control system that comprises one or more thermoelectric elements configured to actively heat or cool at least a portion of the chamber, and one or more power storage elements. The temperature control system also comprises circuitry configured to control an operation of the one or more thermoelectric elements to heat or cool at least a portion of the chamber to a predetermined temperature or temperature range, the circuitry further configured to wirelessly communicate with a cloud-based data storage system or a remote electronic device. An electronic display screen is disposed on one or both of the container body and the lid, the display screen configured to selectively display shipping information for the portable cooler container.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIGS. **1A-1D** are schematic views of one embodiment of a cooler container.
- (2) FIGS. **2A-2B** are schematic partial views of another embodiment of a cooler container.
- (3) FIG. **2C** is a schematic view of another embodiment of a cooler container.
- (4) FIGS. **3A-3C** are schematic partial views of another embodiment of a cooler container.
- (5) FIGS. **4A-4C** are schematic partial views of another embodiment of a cooler container.
- (6) FIGS. **5A-5B** are schematic partial views of another embodiment of a cooler container.
- (7) FIGS. **6A-6B** are schematic partial views of another embodiment of a cooler container.
- (8) FIGS. **7A-7B** are schematic partial views of another embodiment of a cooler container.
- (9) FIGS. **8A-8B** are schematic partial views of another embodiment of a cooler container.
- (10) FIGS. **9A-9B** are schematic partial views of another embodiment of a cooler container.
- (11) FIGS. **10A-10B** are schematic partial views of another embodiment of a cooler container.
- (12) FIG. **11A** is a schematic view of another embodiment of a cooler container.
- (13) FIG. **11B** is a schematic view of another embodiment of a cooler container.
- (14) FIGS. **12A-12B** are schematic partial views of another embodiment of a cooler container.
- (15) FIG. **12C** is a schematic view of another embodiment of a cooler container.
- (16) FIGS. **13A-13B** are schematic partial views of another embodiment of a cooler container.
- (17) FIGS. **14A-14B** are schematic partial views of another embodiment of a cooler container.
- (18) FIGS. **15A-15B** are schematic partial views of another embodiment of a cooler container.
- (19) FIGS. **16A-16B** are schematic partial views of another embodiment of a cooler container.
- (20) FIGS. **17A-17B** are schematic partial views of another embodiment of a cooler container.
- (21) FIG. **18A** is a schematic view of a portion of another embodiment of a cooler container.
- (22) FIG. **18B** is a schematic view of a portion of another embodiment of a cooler container.
- (23) FIG. **18C** is a schematic view of one embodiment of a coupling mechanism between the lid and vessel of the cooler container.
- (24) FIG. **18D** is a schematic view of another embodiment of a coupling mechanism between the lid and the vessel of the cooler container.
- (25) FIG. **18E** is a schematic view of one embodiment of a vessel for the cooler container.
- (26) FIG. **18F** is a schematic view of another embodiment of a vessel for the cooler container.
- (27) FIG. **19** is a schematic view of another embodiment of a cooler container.
- (28) FIG. **20** is a schematic front view of another embodiment of a cooler container.
- (29) FIG. **21** is a schematic rear view of the cooler container of FIG. **20**.
- (30) FIG. **22** is a schematic perspective view of the cooler container of FIG. **20**.
- (31) FIG. **23** is a schematic perspective view of the cooler container of FIG. **20**.
- (32) FIG. **24** is a schematic perspective view of the cooler container of FIG. **20**.
- (33) FIG. **25A** is a schematic view of a tray removed from the container.
- (34) FIG. **25B** is a schematic view of an interchangeable tray system for use with the container.
- (35) FIG. **25C** is a schematic top view of one embodiment of a tray for use in the container of FIG. **20**.
- (36) FIG. **25D** is a schematic top view of another embodiment of a tray for use in the container of FIG. **20**.
- (37) FIG. **26** is a schematic bottom view of the cooler container of FIG. **20**.
- (38) FIG. **27** is a schematic cross-sectional view of the cooler container of FIG. **20** with the tray disposed in the container.
- (39) FIG. **28** is a schematic view of the container in an open position with one or more lighting elements.
- (40) FIGS. **29A-29C** are schematic views of a graphical user interface for use with the container.
- (41) FIG. **30** is a schematic view of a visual display of the container.
- (42) FIG. **31** is a schematic view of security features of the container.

- (43) FIG. **32** is a schematic perspective view of another embodiment of a cooler container.
- (44) FIGS. **33A-33B** are schematic side views of various containers of different sizes.
- (45) FIG. **34** is a schematic view a container disposed on a power base.
- (46) FIGS. **35A-35C** are schematic views of a graphical user interface for use with the container.
- (47) FIG. **36** is a schematic view of another embodiment of a cooler container.
- (48) FIG. **37** is a schematic cross-sectional view of the cooler container of FIG. **32**.
- (49) FIG. **38** is a schematic cross-sectional view of the cooler container of FIG. **37** with one fan in operation.
- (50) FIG. **39** is a schematic cross-sectional view of the cooler container of FIG. **37** with another fan in operation.
- (51) FIG. **40** is a schematic block diagram showing communication between the cooler container and a remote electronic device.
- (52) FIG. **41A** shows a schematic perspective view of a cooler container.
- (53) FIG. **41B** is a is a schematic block diagram showing electronics in the cooler container associated with operation of the display screen of the cooler container.
- (54) FIGS. **42A-42B** show block diagrams of a method for operating the cooler container of FIG. **41A**.

DETAILED DESCRIPTION

- (55) FIGS. **1A-1D** show a schematic cross-sectional view of a container system **100** that includes a cooling system **200**. Optionally, the container system **100** has a container vessel **120** that is optionally cylindrical and symmetrical about a longitudinal axis Z, and one of ordinary skill in the art will recognize that the features shown in cross-section in FIGS. **1A-ID** are defined by rotating them about the axis Z to define the features of the container **100** and cooling system **200**.
- (56) The container vessel **120** is optionally a cooler with active temperature control provided by the cooling system **200** to cool the contents of the container vessel **120** and/or maintain the contents of the vessel **120** in a cooled or chilled state. Optionally, the vessel **120** can hold therein one or more (e.g., a plurality of) separate containers (e.g., vials, cartridges, packages, injectors, etc.). Optionally, the one or more (e.g., plurality of) separate containers that can be inserted into the container vessel **120** are medicine containers (e.g., vaccine vials, insulin cartridges, injectors, etc.).
- (57) The container vessel **120** has an outer wall **121** that extends between a proximal end **122** that has an opening **123** and a distal end **124** having a base **125**. The opening **123** is selectively closed by a lid L removably attached to the proximal end **122**. The vessel **120** has an inner wall **126A** and a base wall **126B** that defines an open chamber **126** that can receive and hold contents to be cooled therein (e.g., one or more volumes of liquid, such as one or more vials, cartridges, packages, injectors, etc.). Optionally, the vessel **120** can be made of metal (e.g., stainless steel). In another implementation, the vessel **120** can be made of plastic. In one implementation, the vessel **120** has a cavity **128** (e.g., annular cavity or chamber) between the inner wall **126A** and the outer wall **121**. Optionally, the cavity **128** can be under vacuum. In another implementation, the cavity **128** can be filled with air but not be under vacuum. In still another implementation, the cavity **128** can be filled with a thermally insulative material (e.g., foam). In another implementation, the vessel **120** can exclude a cavity so that the vessel **120** is solid between the inner wall **126A** and the outer wall **121**.
- (58) With continued reference to FIGS. **1A-1D**, the cooling system **200** is optionally implemented in the lid L that releasably closes the opening **123** of the vessel **120** (e.g., lid L can be attached to vessel **120** to closer the opening **123**, and detached or decoupled from the vessel **120** to access the chamber **126** through the opening **123**).
- (59) The cooling system **200** optionally includes a cold side heat sink **210** that faces the chamber **126**, one or more thermoelectric elements (TECs) **220** (such as one or more Peltier elements) that selectively contacts the cold side heat sink **210**, a hot side heat sink **230** in contact with the thermoelectric element **220** and disposed on an opposite side of the TEC **220** from the cold side heat sink **210**, an insulator member **240** disposed between the cold side heat sink **210** and the hot

side heat sink **230**, one or more distal magnets **250** proximate a surface of the insulator **240**, one or more proximal magnets **260** and one or more electromagnets **270** disposed axially between the distal magnets **250** and the proximal magnets **260**. The proximal magnets **260** have an opposite polarity than the distal magnets **250**. The electromagnets **270** are disposed about and connected to the hot side heat sink **230**, which as noted above is attached to the TEC **220**. The cooling system **200** also optionally includes a fan **280** in communication with the hot side heat sink **230** and one or more sealing gaskets **290** disposed between the cold side heat sink **210** and the hot side heat sink **230** and circumferentially about the TEC **220**.

(60) As discussed further below, circuitry and one or more batteries are optionally disposed in or on the vessel **120**. For example, in one implementation, circuitry, sensors and/or batteries are disposed in a cavity in the distal end **124** of the vessel body **120**, such as below the base wall **126B** of the vessel **120**, and can communicate with electrical contacts on the proximal end **122** of the vessel **120** that can contact corresponding electrical contacts (e.g., pogo pins, contact rings) on the lid **L**. In another implementation, the lid **L** can be connected to the proximal end **122** of the vessel **120** via a hinge, and electrical wires can extend through the hinge between the circuitry disposed in the distal end **124** of the vessel **120** and the fan **280** and TEC **220** in the lid **L**. Further discussion of the electronics in the cooling system **200** is provided further below. In another implementation, the circuitry and one or more batteries can be in a removable pack (e.g., De Walt battery pack) that attaches to the distal end **124** of the vessel **120**, where one or more contacts in the removable pack contact one or more contacts on the distal end **124** of the vessel **120**. The one or more contacts on the distal end **124** of the vessel **120** are electrically connected (via one or more wires or one or more intermediate components) with the electrical connections on the proximal **122** of the vessel **120**, or via the hinge, as discussed above, to provide power to the components of the cooling system **200**.

(61) In operation, the one or more electromagnets **270** are operated to have a polarity that is opposite that of the one or more distal magnets **250** and/or the same as the polarity of the one or more proximal magnets **260**, causing the electromagnets **270** to move toward and contact the distal magnets **250**, thereby causing the TEC **220** to contact the cold side heat sink **210** (see FIG. **1C**). The TEC **220** can be operated to draw heat from the chamber **126** via the cold side heat sink **210**, which the TEC **220** transfers to the hot side heat sink **230**. The fan **280** can optionally be operated to dissipate heat from the hot side heat sink **230**, allowing the TEC **220** to draw more heat out of the chamber **126** to thereby cool the chamber **126**. Once the desired temperature is achieved in the chamber **126** (e.g., as sensed by one or more sensors in thermal communication with the chamber **126**), the fan **280** is turned off and the polarity of the one or more electromagnets **270** can be switched (e.g., switched off) so that the electromagnets **270** are repelled from the distal magnets **250** and/or attracted to the proximal magnets **260**, thereby causing the TEC **220** to be spaced apart from (i.e., no longer contact) the cold side heat sink **210** (see FIG. **1D**) within the housing **225**. The separation between the TEC **220** and the cold side heat sink **210** advantageously prevents heat in the hot side heat sink or due to ambient temperature from flowing back to the cold side heat sink, which prolongs the cooled state in the chamber **126**.

(62) FIGS. **2A-2B** schematically illustrate a container system **100B** that includes the cooling system **200B**. The container system **100B** can include the vessel **120** (as described above). Some of the features of the cooling system **200B** are similar to features in the cooling system **200** in FIGS. **1A-ID**. Thus, references numerals used to designate the various components of the cooling system **200B** are identical to those used for identifying the corresponding components of the cooling system **200** in FIGS. **1A-1D**, except that a “B” is added to the numerical identifier. Therefore, the structure and description for the various components of the cooling system **200** in FIGS. **1A-ID** are understood to also apply to the corresponding components of the cooling system **200B** in FIGS. **2A-2B**, except as described below.

(63) The TEC **220B** can optionally be selectively slid into alignment between the cold side heat

sink **210B** and the hot side heat sink **230B**, such that operation of the TEC **220B** draws heat from the chamber **126** via the cold side heat sink **210B** and transfers it to the hot side heat sink **230B**. The fan **280B** is optionally operated to further dissipate heat from the hot side heat sink **230B**, allowing it to draw more heat from the chamber **126** via the TEC **220B**. Optionally, one or more springs **212B** (e.g., coil springs) resiliently couple the cold side heat sink **210B** with the insulator **240B** to maintain an efficient thermal connection between the cold side heat sink **210B** and the TEC **220** when aligned together.

(64) The TEC **220B** can optionally be selectively slid out of alignment between the cold side heat sink **210B** and the hot side heat sink **230B** to thereby disallow heat transfer through the TEC **220B** (e.g., once the desired temperature in the chamber **126** has been achieved). Optionally, the TEC **220B** is slid into a cavity **242B** in the insulator **240B**.

(65) The TEC **220B** can be slid into and out of alignment between the cold side heat sink **210B** and the hot side heat sink **230B** with a number of suitable mechanisms. In one implementation, an electric motor can drive a gear in contact with a gear rack (e.g., rack and pinion), where the TEC **220B** can be attached to the rack that linearly moved via rotation of the gear by the electric motor. In another implementation, a solenoid motor can be attached to TEC **220B** to effect the linear movement of the TEC **220B**. In still another implementation a pneumatic or electromechanical system can actuate movement of a piston attached to the TEC **220B** to effect the linear movement of the TEC **220B**.

(66) FIG. 2C schematically illustrates a portion of a container system **100B'** that includes the cooling system **200B'**. The container system **100B'** can include the vessel **120** (as described above). Some of the features of the cooling system **200B'** are similar to features in the cooling system **200B** in FIGS. 2A-2B. Thus, references numerals used to designate the various components of the cooling system **200B'** are identical to those used for identifying the corresponding components of the cooling system **200B** in FIGS. 2A-2B, except that a "'" is added to the numerical identifier. Therefore, the structure and description for the various components of the cooling system **200B** in FIGS. 2A-2B are understood to also apply to the corresponding components of the cooling system **200B'** in FIG. 2C, except as described below.

(67) The cooling system **200B'** differs from the cooling system **200B** in that the TEC **220B'** is tapered or wedge shaped. An actuator **20A** (e.g., electric motor) is coupled to the TEC **220B'** via a driver **20B**. The actuator **20A** is selectively actuatable to move the TEC **220B'** into and out of engagement (e.g., into and out of contact) with the hot side heat sink **230B'** and the cold side heat sink **210B'** to allow for heat transfer therebetween. Optionally, the hot side heat sink **230B'** and/or the cold side heat sink **210B'** can have a tapered surface that thermally communicates with (e.g., operatively contacts) one or more tapered surfaces (e.g., wedge shaped surfaces) of the TEC **220B'** when the TEC **220B'** is moved into thermal communication (e.g., into contact) with the hot side heat sink **230B'** and the cold side heat sink **210B'**.

(68) FIGS. 3A-3C schematically illustrate a container system **100C** that includes the cooling system **200C**. The container system **100C** can include the vessel **120** (as described above). Some of the features of the cooling system **200C** are similar to features in the cooling system **200B** in FIGS. 2A-2B. Thus, references numerals used to designate the various components of the cooling system **200C** are identical to those used for identifying the corresponding components of the cooling system **200B** in FIGS. 2A-2B, except that a "C" is used instead of a "B". Therefore, the structure and description for the various components of the cooling system **200B** in FIGS. 2A-2B are understood to also apply to the corresponding components of the cooling system **200C** in FIGS. 3A-3C, except as described below.

(69) The cooling system **200C** differs from the cooling system **200B** in that the TEC **220C** is in a fixed position adjacent the hot side heat sink **230C**. The insulator member **240C** has one or more thermal conductors **244C** embedded therein, and the insulator member **240C** can be selectively rotated about an axis (e.g., an axis offset from the axis Z of the vessel **120**) to align at least one of

the thermal conductors **244C** with the TEC **220C** and the cold side heat sink **210C** to allow heat transfer between the chamber **126** and the hot side heat sink **230C**. The insulator member **240C** can also be selectively rotated to move the one or more thermal conductors **244C** out of alignment with the TEC **220C** so that instead an insulating portion **246C** is interposed between the TEC **220C** and the cold side heat sink **210C**, thereby inhibiting (e.g., preventing) heat transfer between the TEC **220C** and the cold side heat sink **210C** to prolong the cooled state in the chamber **126**. With reference to FIGS. **3B-3C**, in one implementation, the insulator member **240C** can be rotated by a motor **248C** (e.g., electric motor) via a pulley cable or band **249C**.

(70) FIGS. **4A-4C** schematically illustrate a container system **100D** that includes the cooling system **200D**. The container system **100D** can include the vessel **120** (as described above). Some of the features of the cooling system **200D** are similar to features in the cooling system **200C** in FIGS. **3A-3C**. Thus, references numerals used to designate the various components of the cooling system **200D** are identical to those used for identifying the corresponding components of the cooling system **200C** in FIGS. **3A-3C**, except that a “D” is used instead of a “C”. Therefore, the structure and description for the various components of the cooling system **200C** in FIGS. **3A-3C** are understood to also apply to the corresponding components of the cooling system **200D** in FIGS. **4A-4C**, except as described below.

(71) The cooling system **200D** differs from the cooling system **200C** in the mechanism for rotating the insulator member **240D**. In particular, the insulator member **240D** has one or more thermal conductors **244D** embedded therein, and the insulator member **240D** can be selectively rotated about an axis (e.g., an axis offset from the axis Z of the vessel **120**) to align at least one of the thermal conductors **244D** with the TEC **220D** and the cold side heat sink **210D** to allow heat transfer between the chamber **126** and the hot side heat sink **230D**. The insulator member **240D** can also be selectively rotated to move the one or more thermal conductors **244D** out of alignment with the TEC **220D** so that instead an insulating portion **246D** is interposed between the TEC **220D** and the cold side heat sink **210D**, thereby inhibiting (e.g., preventing) heat transfer between the TEC **220D** and the cold side heat sink **210D** to prolong the cooled state in the chamber **126**. With reference to FIGS. **4B-4C**, in one implementation, the insulator member **240D** can be rotated by a motor **248D** (e.g., electric motor) via a gear train or geared connection **249D**.

(72) FIGS. **5A-5B** schematically illustrate a container system **100E** that includes the cooling system **200E**. The container system **100E** can include the vessel **120** (as described above). Some of the features of the cooling system **200D** are similar to features in the cooling system **200B** in FIGS. **2A-2B**. Thus, references numerals used to designate the various components of the cooling system **200E** are identical to those used for identifying the corresponding components of the cooling system **200B** in FIGS. **2A-2B**, except that an “E” is used instead of a “B”. Therefore, the structure and description for the various components of the cooling system **200B** in FIGS. **2A-2B** are understood to also apply to the corresponding components of the cooling system **200E** in FIGS. **5A-5B**, except as described below.

(73) An assembly A including the hot side heat sink **230E**, fan **280E**, TEC **220E** and an insulator segment **244E** can optionally be selectively slid relative to the vessel **120** to bring the TEC **220E** into alignment (e.g., contact) between the cold side heat sink **210E** and the hot side heat sink **230E**, such that operation of the TEC **220E** draws heat from the chamber **126** via the cold side heat sink **210E** and transfers it to the hot side heat sink **230E**. The fan **280E** is optionally operated to further dissipate heat from the hot side heat sink **230E**, allowing it to draw more heat from the chamber **126** via the TEC **220E**. Optionally, one or more springs **212E** (e.g., coil springs) resiliently couple the cold side heat sink **210E** with the insulator **240E** to maintain an efficient thermal connection between the cold side heat sink **210E** and the TEC **220E** when aligned together.

(74) The assembly A can optionally be selectively slid to move the TEC **220E** out of alignment (e.g., contact) between the cold side heat sink **210E** and the hot side heat sink **230E**. This causes the insulator segment **244E** to instead be placed in alignment (e.g., contact) between the cold side heat

sink **210E** and the hot side heat sink **230E**, which disallows heat transfer through the TEC **220E** (e.g., once the desired temperature in the chamber **126** has been achieved).

(75) The assembly **A** can be slid with a number of suitable mechanisms. In one implementation, an electric motor can drive a gear in contact with a gear rack (e.g., rack and pinion), where the assembly **A** can be attached to the rack that linearly moves via rotation of the gear by the electric motor. In another implementation, a solenoid motor can be attached to assembly **A** to effect the linear movement of the assembly **A**. In still another implementation a pneumatic or electromechanical system can actuate movement of a piston attached to the assembly **A** to effect the linear movement of the assembly **A**.

(76) FIGS. **6A-6B** schematically illustrate a container system **100F** that includes the cooling system **200F**. The container system **100F** can include the vessel **120** (as described above). Some of the features of the cooling system **200F** are similar to features in the cooling system **200** in FIGS. **1A-1D**. Thus, references numerals used to designate the various components of the cooling system **200F** are identical to those used for identifying the corresponding components of the cooling system **200** in FIGS. **1A-1D**, except that a “G” is added to the numerical identifiers. Therefore, the structure and description for the various components of the cooling system **200** in FIGS. **1A-1D** are understood to also apply to the corresponding components of the cooling system **200F** in FIGS. **6A-6B**, except as described below.

(77) As shown in FIGS. **6A-6B**, the hot side heat sink **230F** is in contact with the TEC **220F**. One or more springs **212F** (e.g., coil springs) can be disposed between the hot side heat sink **230F** and the insulator member **240F**. The one or more springs **212F** exert a (bias) force on the hot side heat sink **230F** to bias it toward contact with the insulator member **240F**. One or more expandable bladders **250F** are disposed between the insulator member **240F** and the hot side heat sink **230F**.

(78) When the one or more expandable bladders **250F** are in a collapsed state (see FIG. **6A**), the one or more springs **212F** draw the hot side heat sink **230F** toward the insulator member **240F** so that the TEC **220F** contacts the cold side heat sink **210F**. The TEC **220F** can be operated to draw heat out of the chamber **126** via the cold side heat sink **210F**, which is then transferred via the TEC **220F** to the hot side heat sink **230F**. Optionally, the fan **280F** can be operated to dissipate heat from the hot side heat sink **230F**, allowing the hot side heat sink **230F** to draw additional heat from the chamber **126** via the contact between the cold side heat sink **210F**, the TEC **220F** and the hot side heat sink **230F**. Accordingly, with the one or more expandable bladders **250F** in the collapsed state, the cooling system **200F** can be operated to draw heat from the chamber **126** to cool the chamber to a predetermined temperature or temperature range.

(79) When the one or more expandable bladders **250F** are in an expanded state (see FIG. **6B**), they can exert a force on the hot side heat sink **230F** in a direction opposite to the bias force of the one or more springs **212F**, causing the hot side heat sink **230F** to separate from (e.g., lift from) the insulator member **240F**. Such separation between the hot side heat sink **230F** and the insulator member **240F** also causes the TEC **220F** to become spaced apart from the cold side heat sink **210F**, inhibiting (e.g., preventing) heat transfer between the cold side heat sink **210F** and the TEC **220F**. Accordingly, once the predetermined temperature or temperature range has been achieved in the chamber **126**, the one or more expandable bladders **250F** can be transitioned to the expanded state to thermally disconnect the cold side heat sink **210F** from the TEC **220F** to thereby maintain the chamber **126** in a prolonged cooled state.

(80) In one implementation, the one or more expandable bladders **250F** form part of a pneumatic system (e.g., having a pump, one or more valves, and/or a gas reservoir) that selectively fills the bladders **250F** with a gas to move the bladders **250F** to the expanded state and selectively empties the one or more expandable bladders **250F** to move the bladders **250F** to the collapsed state.

(81) In another implementation, the one or more expandable bladders **250F** form part of a hydraulic system (e.g., having a pump, one or more valves, and/or a liquid reservoir) that selectively fills the bladders **250F** with a liquid to move the bladders **250F** to the expanded state and selectively

empties the one or more expandable bladders **250F** to move the bladders **250F** to the collapsed state.

(82) FIGS. 7A-7B schematically illustrate a container system **100G** that includes the cooling system **200G**. The container system **100G** can include the vessel **120** (as described above). Some of the features of the cooling system **200G** are similar to features in the cooling system **200F** in FIGS. 6A-6B. Thus, references numerals used to designate the various components of the cooling system **200G** are identical to those used for identifying the corresponding components of the cooling system **200F** in FIGS. 6A-6B, except that a “G” is used instead of an “F”. Therefore, the structure and description for the various components of the cooling system **200F** in FIGS. 6A-6B are understood to also apply to the corresponding components of the cooling system **200G** in FIGS. 7A-7B, except as described below.

(83) The cooling system **200G** differs from the cooling system **200F** in the position of the one or more springs **212G** and the one or more expandable bladders **250G**. As shown in FIGS. 7A-7B, the one or more springs **212G** (e.g., coil springs) can be disposed between the cold side heat sink **210G** and the insulator member **240G**. The one or more springs **212G** exert a (bias) force on the cold side heat sink **210G** to bias it toward contact with the insulator member **240G**. The one or more expandable bladders **250G** are disposed between the insulator member **240G** and the cold side heat sink **230G**.

(84) When the one or more expandable bladders **250G** are in a collapsed state (see FIG. 7A), the one or more springs **212G** draw the cold side heat sink **230G** (up) toward the insulator member **240G** so that the TEC **220G** contacts the cold side heat sink **210G**. The TEC **220G** can be operated to draw heat out of the chamber **126** via the cold side heat sink **210G**, which is then transferred via the TEC **220G** to the hot side heat sink **230G**. Optionally, the fan **280G** can be operated to dissipate heat from the hot side heat sink **230G**, allowing the hot side heat sink **230G** to draw additional heat from the chamber **126** via the contact between the cold side heat sink **210G**, the TEC **220G** and the hot side heat sink **230G**. Accordingly, with the one or more expandable bladders **250G** in the collapsed state, the cooling system **200G** can be operated to draw heat from the chamber **126** to cool the chamber to a predetermined temperature or temperature range.

(85) When the one or more expandable bladders **250G** are in an expanded state (see FIG. 7B), they can exert a force on the cold side heat sink **210G** in a direction opposite to the bias force of the one or more springs **212G**, causing the cold side heat sink **210G** to separate from (e.g., move down relative to) the insulator member **240G**. Such separation between the cold side heat sink **210G** and the insulator member **240G** also causes the TEC **220G** to become spaced apart from the cold side heat sink **210G**, inhibiting (e.g., preventing) heat transfer between the cold side heat sink **210G** and the TEC **220G**. Accordingly, once the predetermined temperature or temperature range has been achieved in the chamber **126**, the one or more expandable bladders **250G** can be transitioned to the expanded state to thermally disconnect the cold side heat sink **210G** from the TEC **220G** to thereby maintain the chamber **126** in a prolonged cooled state.

(86) In one implementation, the one or more expandable bladders **250G** form part of a pneumatic system (e.g., having a pump, one or more valves, and/or a gas reservoir) that selectively fills the bladders **250G** with a gas to move the bladders **250G** to the expanded state and selectively empties the one or more expandable bladders **250G** to move the bladders **250G** to the collapsed state.

(87) In another implementation, the one or more expandable bladders **250G** form part of a hydraulic system (e.g., having a pump, one or more valves, and/or a liquid reservoir) that selectively fills the bladders **250G** with a liquid to move the bladders **250G** to the expanded state and selectively empties the one or more expandable bladders **250G** to move the bladders **250G** to the collapsed state.

(88) FIGS. 8A-8B schematically illustrate a container system **100H** that includes the cooling system **200H**. The container system **100H** can include the vessel **120** (as described above). Some of the features of the cooling system **200H** are similar to features in the cooling system **200F** in FIGS.

6A-6B. Thus, reference numerals used to designate the various components of the cooling system **200H** are identical to those used for identifying the corresponding components of the cooling system **200F** in FIGS. **6A-6B**, except that an “H” is used instead of an “F”. Therefore, the structure and description for the various components of the cooling system **200F** in FIGS. **6A-6B** are understood to also apply to the corresponding components of the cooling system **200H** in FIGS. **8A-8B**, except as described below.

(89) The cooling system **200H** differs from the cooling system **200F** in that one or more expandable bladders **255H** are included instead of the one or more springs **212F** to provide a force in a direction opposite to the force exerted by the one or more expandable bladders **250H**. As shown in FIGS. **8A-8B**, the one or more expandable bladders **255H** are disposed between a housing **225H** and a portion of the hot side heat sink **230H**, and one or more expandable bladders **250H** are disposed between the insulator member **240H** and the hot side heat sink **230H**. Optionally, the one or more expandable bladders **250H** are in fluid communication with the one or more expandable bladders **255H**, and the fluid is moved between the two expandable bladders **250H**, **255H**. That is, when the one or more expandable bladders **250H** are in the expanded state, the one or more expandable bladders **255H** are in the collapsed state, and when the expandable bladders **250H** are in the collapsed state, the expandable bladders **255H** are in the expanded state.

(90) When the one or more expandable bladders **250H** are in a collapsed state (see FIG. **8A**), the one or more expandable bladders **255H** are in the expanded state and exert a force on the hot side heat sink **230H** toward the insulator member **240H** so that the TEC **220H** contacts the cold side heat sink **210H**. The TEC **220H** can be operated to draw heat out of the chamber **126** via the cold side heat sink **210H**, which is then transferred via the TEC **220H** to the hot side heat sink **230H**. Optionally, the fan **280H** can be operated to dissipate heat from the hot side heat sink **230H**, allowing the hot side heat sink **230H** to draw additional heat from the chamber **126** via the contact between the cold side heat sink **210H**, the TEC **220H** and the hot side heat sink **230H**. Accordingly, with the one or more expandable bladders **250H** in the collapsed state, the cooling system **200H** can be operated to draw heat from the chamber **126** to cool the chamber to a predetermined temperature or temperature range.

(91) When the one or more expandable bladders **250H** are in an expanded state (see FIG. **8B**), the one or more expandable bladders **255H** are in a collapsed state. The expanded state of the expandable bladders **250H** exerts a force on the hot side heat sink **230H** that causes the hot side heat sink **230H** to separate from (e.g., lift from) the insulator member **240H**. Such separation between the hot side heat sink **230H** and the insulator member **240H** also causes the TEC **220H** to become spaced apart from (e.g., lift from) the cold side heat sink **210H**, thereby thermally disconnecting (e.g., inhibiting heat transfer between) the cold side heat sink **210H** and the TEC **220H**. Accordingly, once the predetermined temperature or temperature range has been achieved in the chamber **126**, the one or more expandable bladders **250H** can be transitioned to the expanded state (e.g., by transferring the fluid from the expandable bladders **255H** to the expandable bladders **250H**) to thermally disconnect the cold side heat sink **210H** from the TEC **220H** to thereby maintain the chamber **126** in a prolonged cooled state.

(92) In one implementation, the one or more expandable bladders **250H**, **255H** form part of a pneumatic system (e.g., having a pump, one or more valves, and/or a gas reservoir) that selectively fills and empties the bladders **250H**, **255H** with a gas to move them between an expanded and a collapsed state.

(93) In one implementation, the one or more expandable bladders **250H**, **255H** form part of a hydraulic system (e.g., having a pump, one or more valves, and/or a liquid reservoir) that selectively fills and empties the bladders **250H**, **255H** with a liquid to move them between an expanded and a collapsed state.

(94) FIGS. **9A-9B** schematically illustrate a container system **100I** that includes the cooling system **200I**. The container system **100I** can include the vessel **120** (as described above). Some of the

features of the cooling system **200I** are similar to features in the cooling system **200G** in FIGS. 7A-7B. Thus, references numerals used to designate the various components of the cooling system **200I** are identical to those used for identifying the corresponding components of the cooling system **200G** in FIGS. 7A-7B, except that an “I” is used instead of a “G”. Therefore, the structure and description for the various components of the cooling system **200G** in FIGS. 7A-7B are understood to also apply to the corresponding components of the cooling system **200I** in FIGS. 9A-9B, except as described below.

(95) The cooling system **200I** differs from the cooling system **200G** in that the one or more rotatable cams **250I** are used instead of one or more expandable bladders **250G**. As shown in FIGS. 9A-9B, the one or more springs **212I** (e.g., coil springs) can be disposed between the cold side heat sink **210I** and the insulator member **240I**. The one or more springs **212I** exert a (bias) force on the cold side heat sink **210I** to bias it toward contact with the insulator member **240I**. The one or more rotatable cams **250I** are rotatably coupled to the insulator member **240I** and rotatable to selectively contact a proximal surface of the cold side heat sink **230I**.

(96) In a cooling state (see FIG. 9A), the rotatable cams **250I** are not in contact with the cold side heat sink **210I**, such that the one or more springs **212I** bias the cold side heat sink **210I** into contact with the TEC **220I**, thereby allowing heat transfer therebetween. The TEC **220I** can be operated to draw heat out of the chamber **126** via the cold side heat sink **210I**, which is then transferred via the TEC **220I** to the hot side heat sink **230I**. Optionally, the fan **280I** can be operated to dissipate heat from the hot side heat sink **230I**, allowing the hot side heat sink **230I** to draw additional heat from the chamber **126** via the contact between the cold side heat sink **210I**, the TEC **220I** and the hot side heat sink **230I**. Accordingly, with the one or more rotatable cams **250I** in a retracted state, the cooling system **200I** can be operated to draw heat from the chamber **126** to cool the chamber to a predetermined temperature or temperature range.

(97) When the one or more rotatable cams **250I** are moved to the deployed state (see FIG. 9B), the cams **250I** bear against the cold side heat sink **210I**, overcoming the bias force of the springs **212I**. In the deployed state, the one or more cams **250I** exert a force on the cold side heat sink **210I** that causes the cold side heat sink **210I** to separate from (e.g., move down relative to) the insulator member **240I**. Such separation between the cold side heat sink **210I** and the insulator member **240I** also causes the cold side heat sink **210I** to become spaced apart from (e.g., move down relative to) the TEC **220I**, thereby thermally disconnecting (e.g., inhibiting heat transfer between) the cold side heat sink **210I** and the TEC **220I**. Accordingly, once the predetermined temperature or temperature range has been achieved in the chamber **126**, the one or more rotatable cams **250I** can be moved to the deployed state to thermally disconnect the cold side heat sink **210I** from the TEC **220I** to thereby maintain the chamber **126** in a prolonged cooled state.

(98) FIGS. 10A-10B schematically illustrate a container system **100J** that includes the cooling system **200J**. The container system **100J** can include the vessel **120** (as described above). Some of the features of the cooling system **200J** are similar to features in the cooling system **200I** in FIGS. 9A-9B. Thus, references numerals used to designate the various components of the cooling system **200J** are identical to those used for identifying the corresponding components of the cooling system **200I** in FIGS. 9A-9B, except that an “J” is used instead of an “I”. Therefore, the structure and description for the various components of the cooling system **200I** in FIGS. 9A-9B are understood to also apply to the corresponding components of the cooling system **200J** in FIGS. 10A-10B, except as described below.

(99) The cooling system **200J** differs from the cooling system **200I** in the location of the one or more springs **212J** and the one or more cams **250J**. As shown in FIGS. 10A-10B, the one or more springs **212J** are disposed between the insulator member **240J** and the hot side heat sink **230J** and exert a bias force between the two biasing the hot side heat sink **230J** down toward contact with the insulator member **240J**. Such bias force also biases the TEC **220J** (which is attached to or in contact with the hot side heat sink **230J**) into contact with the cold side heat sink **210J**.

(100) When the one or more rotatable cams **250J** are in a retracted state (see FIG. **10A**), the cams **250J** allow the TEC **220J** to contact the cold side heat sink **210J**. The TEC **220J** can be operated to draw heat out of the chamber **126** via the cold side heat sink **210J**, which is then transferred via the TEC **220J** to the hot side heat sink **230J**. Optionally, the fan **280J** can be operated to dissipate heat from the hot side heat sink **230J**, allowing the hot side heat sink **230J** to draw additional heat from the chamber **126** via the contact between the cold side heat sink **210J**, the TEC **220J** and the hot side heat sink **230J**. Accordingly, with the one or more rotatable cams **250J** in a retracted state, the cooling system **200J** can be operated to draw heat from the chamber **126** to cool the chamber to a predetermined temperature or temperature range.

(101) When the one or more rotatable cams **250J** are moved to the deployed state (see FIG. **10B**), the cams **250J** bear against the hot side heat sink **230J**, overcoming the bias force of the springs **212J**. In the deployed state, the one or more cams **250J** exert a force on the hot side heat sink **230J** that causes the hot side heat sink **230J** to separate from (e.g., lift from) the insulator member **240J**. Such separation also causes the TEC **220J** (attached to the hot side heat sink **230J**) to become spaced apart from (e.g., lift from) the cold side heat sink **210J**, thereby thermally disconnecting (e.g., inhibiting heat transfer between) the cold side heat sink **210J** and the TEC **220J**. Accordingly, once the predetermined temperature or temperature range has been achieved in the chamber **126**, the one or more rotatable cams **250J** can be moved to the deployed state to thermally disconnect the cold side heat sink **210J** from the TEC **220J** to thereby maintain the chamber **126** in a prolonged cooled state.

(102) FIG. **11A** schematically illustrates a container system **100K** that includes the cooling system **200K**. The container system **100K** can include the vessel **120** (as described above) removably sealed by a lid **L'**. Some of the features of the cooling system **200K** are similar to features in the cooling system **200** in FIGS. **1A-1D**. Thus, reference numerals used to designate the various components of the cooling system **200K** are similar to those used for identifying the corresponding components of the cooling system **200** in FIGS. **1A-ID**, except that an "K" is used. Therefore, the structure and description for said similar components of the cooling system **200** in FIGS. **1A-ID** are understood to also apply to the corresponding components of the cooling system **200K** in FIG. **11A**, except as described below.

(103) With reference to FIG. **11A**, the vessel **120** optionally has a cavity **128** (e.g., annular cavity or chamber) between the inner wall **126A** and the outer wall **121**. The cavity **128** can be under vacuum, so that the vessel **120** is vacuum sealed. The lid **L'** that removably seals the vessel **120** is optionally also a vacuum sealed lid. The vacuum sealed vessel **120** and/or lid **L'** advantageously inhibits heat transfer therethrough, thereby inhibiting a passive change in temperature in the chamber **126** when the lid **L'** is attached to the vessel **120** (e.g., via passive loss of cooling through the wall of the vessel **120** and/or lid **L'**).

(104) The cooling system **200K** includes a hot side heat sink **230K** in thermal communication with the thermoelectric element (TEC) (e.g., Peltier element) **220K**, so that the heat sink **230K** can draw heat away from the TEC **220K**. Optionally, a fan **280K** can be in thermal communication with the hot side heat sink **230K** and be selectively operable to further dissipate heat from the hot side heat sink **230K**, thereby allowing the heat sink **230K** to further draw heat from the TEC **220K**.

(105) The TEC **220K** is in thermal communication with a cold side heat sink **210K**, which is in turn in thermal communication with the chamber **126** in the vessel **120**. The cold side heat sink **210K** optionally includes a flow path **214K** that extends from an opening **132K** in the lid **L'** adjacent the chamber **126** to an opening **134K** in the lid **L'** adjacent the chamber **126**. In one implementation, the opening **132K** is optionally located generally at a center of the lid **L'**, as shown in FIG. **11A**. In one implementation, the opening **134K** is optionally located in the lid **L'** at a location proximate the inner wall **126A** of the vessel **120** when the lid **L'** is attached to the vessel **120**. Optionally, the cold side heat sink **210K** includes a fan **216K** disposed along the flow path **214K** between the openings **132K**, **134K**. As shown in FIG. **11A**, at least a portion of the flow path **214K** is in thermal

communication with the TEC **220K** (e.g., with a cold side of the TEC).

(106) In operation, air in the chamber **126** enters the flow path **214K** via the opening **132K** and flows through the flow path **214K** so that it passes through the portion of the flow path **214K** that is proximate the TEC **220K**, where the TEC **220K** is selectively operated to cool (e.g., reduce the temperature of) the air flow passing therein. The cooled airflow continues to flow through the flow path **214K** and exits the flow path **214K** at opening **134K** where it enters the chamber **126**. Optionally, the fan **216K** is operable to draw (e.g., cause or facilitate) the flow of air through the flow path **214K**.

(107) Though FIG. **11A** shows the cooling system **200** disposed on a side of the vessel **120**, one of skill in the art will recognize that the cooling system **200** can be disposed in other suitable locations (e.g., on the bottom of the vessel **120**, on top of the lid **L'**, in a separate module attachable to the top of the lid **L'**, etc.) and that such implementations are contemplated by the invention.

(108) FIG. **11B** schematically illustrates a container system **100K'** that includes the cooling system **200K'**. The container system **100K'** can include the vessel **120** (as described above). Some of the features of the cooling system **200K'** are similar to features in the cooling system **200K** in FIG. **11A**. Thus, reference numerals used to designate the various components of the cooling system **200K'** are similar to those used for identifying the corresponding components of the cooling system **200K** in FIG. **11A**, except that an "'" is used. Therefore, the structure and description for said similar components of the cooling system **200K** in FIG. **11A** are understood to also apply to the corresponding components of the cooling system **200K'** in FIG. **11B**, except as described below.

(109) The container system **100K'** is optionally a self-chilled container (e.g. self-chilled water container, such as a water bottle). The cooling system **200K'** differs from the cooling system **200K** in that a liquid is used as a cooling medium that is circulated through the body of the vessel **120**. A conduit **134K'** can deliver chilled liquid to the body of the vessel **120**, and a conduit **132K'** can remove a warm liquid from the body of the vessel **120**. In the body of the vessel **120**, the chilled liquid can absorb energy from one or more walls of the vessel **120** (e.g., one or more walls that define the chamber **126**) of a liquid in the chamber **126**, and the heated liquid can exit the body of the vessel **120** via conduit **132K'**. In this manner, one or more surfaces of the body of the vessel **120** (e.g., of the chamber **126**) are maintained in the cooled state. Though not shown, the conduits **132K'**, **134K'** connect to a cooling system, such as one having a TEC **220K** in contact with a hot side heat sink **230K**, as described above for container system **100K**.

(110) FIGS. **12A-12B** schematically illustrate a container system **100L** that includes the cooling system **200L**. The container system **100L** can include the vessel **120** (as described above). Some of the features of the cooling system **200L**, which optionally serves as part of the lid **L** that selectively seals the vessel **120**, are similar to features in the cooling system **200** in FIGS. **1A-1D**. Thus, reference numerals used to designate the various components of the cooling system **200L** are similar to those used for identifying the corresponding components of the cooling system **200** in FIGS. **1A-1D**, except that an "L" is used. Therefore, the structure and description for said similar components of the cooling system **200** in FIGS. **1A-1D** are understood to also apply to the corresponding components of the cooling system **200L** in FIGS. **12A-12B**, except as described below.

(111) With reference to FIGS. **12A-12B**, the cooling system **200L** can optionally include a cavity **214L** disposed between the thermoelectric element (TEC) **220L** and the cold side heat sink **210L**. The cooling system **200L** can optionally include a pump **216L** (e.g., a peristaltic pump) in fluid communication with the cavity **214L** and with a reservoir **213L**. The pump **216L** is operable to move a conductive fluid **217L** (e.g., a conductive liquid), such as a volume of conductive fluid **217L**, between the reservoir **213L** and the cavity **214L**. Optionally, the conductive fluid **217L** can be mercury; however, the conductive fluid **217L** can be other suitable liquids.

(112) In operation, when the cooling system **200L** is operated in a cooling stage, the pump **216L** is selectively operable to pump the conductive fluid **217L** into the cavity **214L** (e.g., to fill the cavity

214L), thereby allowing heat transfer between the cold side heat sink **210L** and the TEC **220L** (e.g., allowing the TEC **220L** to be operated to draw heat from the cold side heat sink **210L** and transfer it to the hot side heat sink **230L**). Optionally, the fan **280L** is selectively operable to dissipate heat from the hot side heat sink **230L**, thereby allowing the TEC **220L** to draw further heat from the chamber **126** via the cold side heat sink **210L** and the conductive fluid **217L**.

(113) With reference to FIG. **12A**, when the cooling system **200L** is operated in an insulating state, the pump **216L** is selectively operated to remove (e.g., drain) the conductive fluid **217L** from the cavity **214L** (e.g., by moving the conductive fluid **217L** into the reservoir **213L**), thereby leaving the cavity **214L** unfilled (e.g., empty). Such removal (e.g., complete removal) of the conductive fluid **217L** from the cavity **214L** thermally disconnects the cold side heat sink **210L** from the TEC **220L**, thereby inhibiting (e.g., preventing) heat transfer between the TEC **220L** and the chamber **126** via the cold side heat sink **210L**, which advantageously prevents heat in the hot side heat sink **230L** or due to ambient temperature from flowing back to the cold side heat sink **210L**, thereby prolonging the cooled state in the chamber **126**.

(114) FIG. **12C** schematically illustrate a container system **100L'** that includes the cooling system **200L'**. The container system **100L'** can include the vessel **120** (as described above). Some of the features of the cooling system **200L'** are similar to features in the cooling system **200L** in FIGS. **12A-12B**. Thus, references numerals used to designate the various components of the cooling system **200L'** are similar to those used for identifying the corresponding components of the cooling system **200L** in FIGS. **12A-12B**, except that an "'" is used. Therefore, the structure and description for said similar components of the cooling system **200L** in FIGS. **12A-12B** are understood to also apply to the corresponding components of the cooling system **200L'** in FIG. **12C**, except as described below.

(115) The cooling system **200L'** differs from the cooling system **200L** in that a heat pipe **132L'** is used to connect the hot side heat sink **230L'** to the cold side heat sink **210L'**. The heat pipe **132L'** can be selectively turned on and off. Optionally, the heat pipe **132L'** can include a phase change material (PCM). Optionally, the heat pipe **132L'** can be turned off by removing the working fluid from inside the heat pipe **132L'**, and turned on by inserting or injecting the working fluid in the heat pipe **132L'**. For example, the TEC **220L'**, when in operation, can freeze the liquid in the heat pipe **132L'**, to thereby provide a thermal break within the heat pipe **132L'**, disconnecting the chamber of the vessel **120** from the TEC **220L'** that is operated to cool the chamber. When the TEC **220L'** is not in operation, the liquid in the heat pipe **132L'** can flow along the length of the heat pipe **132L'**. For example, the fluid can flow within the heat pipe **132L'** into thermal contact with a cold side of the TEC **220L'**, which can cool the liquid, the liquid can then flow to the hot side of the heat pipe **132L'** and draw heat away from the chamber of the vessel **120** which heats such liquid, and the heated liquid can then again flow to the opposite end of the heat pipe **132L'** where the TEC **220L'** can again remove heat from it to cool the liquid before it again flows back to the other end of the heat pipe **132L'** to draw more heat from the chamber.

(116) FIGS. **13A-13B** schematically illustrate a container system **100M** that includes the cooling system **200M**. The container system **100M** can include the vessel **120** (as described above). Some of the features of the cooling system **200M**, which optionally serves as part of the lid **L** that selectively seals the vessel **120**, are similar to features in the cooling system **200** in FIGS. **1A-1D**. Thus, references numerals used to designate the various components of the cooling system **200M** are similar to those used for identifying the corresponding components of the cooling system **200** in FIGS. **1A-1D**, except that an "M" is used. Therefore, the structure and description for said similar components of the cooling system **200** in FIGS. **1A-1D** are understood to also apply to the corresponding components of the cooling system **200M** in FIGS. **13A-13B**, except as described below.

(117) With reference to FIGS. **13A-13B**, the cooling system **200M** can include a cold side heat sink **210M** in thermal communication with a thermoelectric element (TEC) **220M** and can selectively be

in thermal communication with the chamber **126** of the vessel. Optionally, the cooling system **200** can include a fan **216M** selectively operable to draw air from the chamber **126** into contact with the cold side heat sink **210M**. Optionally, cooling system **200M** can include an insulator member **246M** selectively movable (e.g., slidable) between one or more positions. As shown in FIGS. **13A-13B**, the insulator member **246M** can be disposed adjacent or in communication with the chamber **126**. (118) With reference to FIG. **13A**, when the cooling system **200M** is operated in a cooling state, the insulator member **246M** is disposed at least partially apart (e.g., laterally apart) relative to the cold side heat sink **210M** and fan **216M**. The TEC **220M** is selectively operated to draw heat from the cold side heat sink **210M** and transfer it to the hot side heat sink **230M**. Optionally, a fan **280M** is selectively operable to dissipate heat from the hot side heat sink **230M**, thereby allowing the TEC **220M** to draw further heat from the chamber **126** via the cold side heat sink **210M**.

(119) With reference to FIG. **13B**, when the cooling system **200M** is operated in an insulating stage, the insulator member **246M** is moved (e.g., slid) into a position adjacent to the cold side heat sink **210M** so as to be disposed between the cold side heat sink **210M** and the chamber **126**, thereby blocking air flow to the cold side heat sink **210M** (e.g., thermally disconnecting the cold side heat sink **210M** from the chamber **126**) to thereby inhibit heat transfer to and from the chamber **126** (e.g., to maintain the chamber **126** in an insulated state).

(120) The insulator member **246M** can be moved between the position in the cooling state (see FIG. **13A**) and the position in the insulating stage (see FIG. **13B**) using any suitable mechanism (e.g., electric motor, solenoid motor, a pneumatic or electromechanical system actuating a piston attached to the insulator member **246M**, etc.). Though the insulator member **246M** is shown in FIGS. **13A-13B** as sliding between said positions, in another implementation, the insulator member **246M** can rotate between the cooling stage position and the insulating stage position.

(121) FIGS. **14A-14B** schematically illustrate a container system **100N** that includes the cooling system **200N**. The container system **100N** can include the vessel **120** (as described above). Some of the features of the cooling system **200N**, which optionally serves as part of the lid **L** that selectively seals the vessel **120**, are similar to features in the cooling system **200M** in FIGS. **13A-13B**. Thus, references numerals used to designate the various components of the cooling system **200N** are similar to those used for identifying the corresponding components of the cooling system **200M** in FIGS. **13A-13B**, except that an “N” is used. Therefore, the structure and description for said similar components of the cooling system **200M** in FIGS. **13A-13B** are understood to also apply to the corresponding components of the cooling system **200N** in FIGS. **14A-14B**, except as described below.

(122) With reference to FIGS. **14A-14B**, the cooling system **200N** can include a cold side heat sink **210N** in thermal communication with a thermoelectric element (TEC) **220N** and can selectively be in thermal communication with the chamber **126** of the vessel **120**. Optionally, the cooling system **200N** can include a fan **216N** selectively operable to draw air from the chamber **126** into contact with the cold side heat sink **210N** via openings **132N**, **134N** and cavities or chambers **213N**, **214N**. Optionally, cooling system **200N** can include insulator members **246N**, **247N** selectively movable (e.g., pivotable) between one or more positions relative to the openings **134N**, **132N**, respectively. As shown in FIGS. **14A-14B**, the insulator member **246N** can be disposed adjacent or in communication with the chamber **126** and be movable to selectively allow and disallow airflow through the opening **134N**, and the insulator member **247N** can be disposed in the chamber **214N** and be movable to selectively allow and disallow airflow through the opening **132N**.

(123) With reference to FIG. **14A**, when the cooling system **200N** is operated in a cooling state, the insulator members **246N**, **247N** are disposed at least partially apart from the openings **134N**, **132N**, respectively, allowing air flow from the chamber **126** through the openings **132N**, **134N** and cavities **213N**, **214N**. Optionally, the fan **216N** can be operated to draw said airflow from the chamber **126**, through the opening **132N** into the chamber **214N** and over the cold side heat sink **210N**, then through the chamber **213N** and opening **134N** and back to the chamber **126**. The TEC

220N is selectively operated to draw heat from the cold side heat sink **210N** and transfer it to the hot side heat sink **230N**. Optionally, a fan **280N** is selectively operable to dissipate heat from the hot side heat sink **230N**, thereby allowing the TEC **220N** to draw further heat from the chamber **126** via the cold side heat sink **210N**.

(124) With reference to FIG. **14B**, when the cooling system **200N** is operated in an insulating stage, the insulator members **246N**, **247N** are moved (e.g., pivoted) into a position adjacent to the openings **134N**, **132N**, respectively to close said openings, thereby blocking air flow to the cold side heat sink **210N** (e.g., thermally disconnecting the cold side heat sink **210N** from the chamber **126**) to thereby inhibit heat transfer to and from the chamber **126** (e.g., to maintain the chamber **126** in an insulated state).

(125) The insulator members **246N**, **247N** can be moved between the position in the cooling state (see FIG. **14A**) and the position in the insulating stage (see FIG. **14B**) using any suitable mechanism (e.g., electric motor, solenoid motor, etc.). Optionally, the insulator members **246N**, **247N** are spring loaded into the closed position (e.g., adjacent the openings **134N**, **132N**), such that the insulator members **246N**, **247N** are pivoted to the open position (see FIG. **14A**) automatically with an increase in air pressure generated by the operation of the fan **216N**. Though the insulator members **246N**, **247N** are shown in FIGS. **14A-14B** as pivoting between said positions, in another implementation, the insulator members **246N**, **247N** can slide or translate between the cooling stage position and the insulating stage position.

(126) FIGS. **15A-15B** schematically illustrate a container system **100P** that includes the cooling system **200P**. The container system **100P** can include the vessel **120** (as described above). Some of the features of the cooling system **200P**, which optionally serves as part of the lid **L** that selectively seals the vessel **120**, are similar to features in the cooling system **200M** in FIGS. **13A-13B**. Thus, references numerals used to designate the various components of the cooling system **200P** are similar to those used for identifying the corresponding components of the cooling system **200M** in FIGS. **13A-13B**, except that an “P” is used. Therefore, the structure and description for said similar components of the cooling system **200M** in FIGS. **13A-13B** are understood to also apply to the corresponding components of the cooling system **200P** in FIGS. **15A-15B**, except as described below.

(127) With reference to FIGS. **15A-15B**, the cooling system **200P** can include a cold side heat sink **210P** in thermal communication with a thermoelectric element (TEC) **220P** and can selectively be in thermal communication with the chamber **126** of the vessel **120**. Optionally, the cooling system **200P** can include a fan **216P** selectively operable to draw air from the chamber **126** into contact with the cold side heat sink **210P**. Optionally, cooling system **200P** can include insulator members **246P**, **247P** selectively movable (e.g., slidable) between one or more positions relative to the cold side heat sink **210P**.

(128) With reference to FIG. **15A**, when the cooling system **200P** is operated in a cooling state, the insulator members **246P**, **247P** are disposed at least partially apart from the cold side heat sink **210P**, allowing air flow from the chamber **126** to contact (e.g., be cooled by) the cold side heat sink **210P**. Optionally, the fan **216P** can be operated to draw said airflow from the chamber **126** and over the cold side heat sink **210P**. The TEC **220P** is selectively operated to draw heat from the cold side heat sink **210P** and transfer it to the hot side heat sink **230P**. Optionally, a fan **280P** is selectively operable to dissipate heat from the hot side heat sink **230P**, thereby allowing the TEC **220P** to draw further heat from the chamber **126** via the cold side heat sink **210P**.

(129) With reference to FIG. **15B**, when the cooling system **200P** is operated in an insulating stage, the insulator members **246P**, **247P** are moved (e.g., slid) into a position between the cold side heat sink **210P** and the chamber **126**, thereby blocking air flow to the cold side heat sink **210P** (e.g., thermally disconnecting the cold side heat sink **210P** from the chamber **126**) to thereby inhibit heat transfer to and from the chamber **126** (e.g., to maintain the chamber **126** in an insulated state).

(130) The insulator members **246P**, **247P** can be moved between the position in the cooling state

(see FIG. 15A) and the position in the insulating stage (see FIG. 15B) using any suitable mechanism (e.g., electric motor, solenoid motor, etc.). Though the insulator members **246P**, **247P** are shown in FIGS. 15A-15B as sliding between said positions, in another implementation, the insulator members **246P**, **247P** can pivot between the cooling stage position and the insulating stage position.

(131) FIGS. 16A-16B schematically illustrate a container system **100Q** that includes the cooling system **200Q**. The container system **100Q** can include the vessel **120** (as described above). Some of the features of the cooling system **200Q**, which optionally serves as part of the lid **L** that selectively seals the vessel **120**, are similar to features in the cooling system **200M** in FIGS. 13A-13B. Thus, references numerals used to designate the various components of the cooling system **200Q** are similar to those used for identifying the corresponding components of the cooling system **200M** in FIGS. 13A-13B, except that an “Q” is used. Therefore, the structure and description for said similar components of the cooling system **200M** in FIGS. 13A-13B are understood to also apply to the corresponding components of the cooling system **200Q** in FIGS. 16A-16B, except as described below.

(132) With reference to FIGS. 16A-16B, the cooling system **200Q** can include a cold side heat sink **210Q** in thermal communication with a thermoelectric element (TEC) **220Q** and can selectively be in thermal communication with the chamber **126** of the vessel **120**. Optionally, the cooling system **200Q** can include a fan **216Q** selectively operable to draw air from the chamber **126** into contact with the cold side heat sink **210Q**. Optionally, the cooling system **200Q** can include an expandable members **246Q** selectively movable between A deflated state and an expanded state relative to the cold side heat sink **210P**.

(133) With reference to FIG. 16A, when the cooling system **200Q** is operated in a cooling state, the expandable member **246Q** is in the deflated state, allowing air flow from the chamber **126** to contact (e.g., be cooled by) the cold side heat sink **210Q**. Optionally, the fan **216Q** can be operated to draw said airflow from the chamber **126** and over the cold side heat sink **210Q**. The TEC **220Q** is selectively operated to draw heat from the cold side heat sink **210Q** and transfer it to the hot side heat sink **230Q**. Optionally, a fan **280Q** is selectively operable to dissipate heat from the hot side heat sink **230Q**, thereby allowing the TEC **220Q** to draw further heat from the chamber **126** via the cold side heat sink **210Q**.

(134) With reference to FIG. 16B, when the cooling system **200Q** is operated in an insulating stage, the expandable member **246Q** is moved into the expanded state so that the expandable member **246Q** is between the cold side heat sink **210Q** and the chamber **126**, thereby blocking air flow to the cold side heat sink **210Q** (e.g., thermally disconnecting the cold side heat sink **210Q** from the chamber **126**) to thereby inhibit heat transfer to and from the chamber **126** (e.g., to maintain the chamber **126** in an insulated state).

(135) The expandable member **246Q** is optionally disposed or house in a cavity or chamber **242Q** defined in the insulator member **240Q**. Optionally, the expandable member **246Q** is part of a pneumatic system and filled with a gas (e.g., air) to move it into the expanded state. In another implementation, the expandable member **246Q** is part of a hydraulic system and filled with a liquid (e.g., water) to move it into the expanded state.

(136) FIGS. 17A-17B schematically illustrate a container system **100R** that includes the cooling system **200R**. The container system **100R** can include the vessel **120** (as described above). Some of the features of the cooling system **200R**, which optionally serves as part of the lid **L** that selectively seals the vessel **120**, are similar to features in the cooling system **200M** in FIGS. 13A-13B. Thus, references numerals used to designate the various components of the cooling system **200R** are similar to those used for identifying the corresponding components of the cooling system **200M** in FIGS. 13A-13B, except that an “R” is used. Therefore, the structure and description for said similar components of the cooling system **200M** in FIGS. 13A-13B are understood to also apply to the corresponding components of the cooling system **200R** in FIGS. 17A-17B, except as described

below.

(137) With reference to FIGS. **17A-17B**, the cooling system **200R** can include a cold side heat sink **210R** in thermal communication with a thermoelectric element (TEC) **220R** and can selectively be in thermal communication with the chamber **126** of the vessel. Optionally, the cooling system **200** can include a fan **216R** selectively operable to draw air from the chamber **126** into contact with the cold side heat sink **210R**. Optionally, cooling system **200R** can include an insulator element **246R** selectively movable (e.g., pivotable) between one or more positions. As shown in FIGS. **17A-17B**, the insulator element **246R** can be disposed in a cavity or chamber **242R** defined in the insulator member **240R**.

(138) With reference to FIG. **17A**, when the cooling system **200R** is operated in a cooling state, the insulator element **246R** is disposed relative to the cold side heat sink **210R** so as to allow air flow through the chamber **242R** from the chamber **126** to the cold side heat sink **210R**. Optionally, the fan **216R** is selectively operated to draw air from the chamber **126** into contact with the cold side heat sink **210R** (e.g., to cool said air flow and return it to the chamber **126**). The TEC **220R** is selectively operated to draw heat from the cold side heat sink **210R** and transfer it to the hot side heat sink **230R**. Optionally, a fan **280R** is selectively operable to dissipate heat from the hot side heat sink **230R**, thereby allowing the TEC **220R** to draw further heat from the chamber **126** via the cold side heat sink **210R**.

(139) With reference to FIG. **17B**, when the cooling system **200R** is operated in an insulating stage, the insulator element **246R** is moved (e.g., rotated, pivoted) into a position relative to the cold side heat sink **210P** so as to close off the chamber **242R**, thereby blocking air flow from the chamber **126** to the cold side heat sink **210R** (e.g., thermally disconnecting the cold side heat sink **210R** from the chamber **126**) to thereby inhibit heat transfer to and from the chamber **126** (e.g., to maintain the chamber **126** in an insulated state).

(140) The insulator element **246R** can be moved between the position in the cooling state (see FIG. **17A**) and the position in the insulating stage (see FIG. **17B**) using any suitable mechanism (e.g., electric motor, solenoid motor, etc.).

(141) FIG. **18A** is a schematic view of a portion of a cooling system **200S**. The cooling system **200S** is similar to the cooling systems disclosed herein, such as cooling systems **200-200X**, except as described below.

(142) As shown in FIG. **18A**, in the cooling system **200S**, the fan **280S** has air intake **I** that is generally vertical and air exhaust **E** that is generally horizontal, so that the air flows generally horizontally over one or more heat sink surfaces, such as surfaces of the hot side heat sink **230S**.

(143) FIG. **18B** is a schematic view of a portion of a cooling system **200T**. The cooling system **200T** in a cylindrical container **100T** has a fan **280T** that optionally blows air over a heat sink **230T**. Optionally, the cooling system **200T** has a heat pipe **132T** in thermal communication with another portion of the container **100T** via end portion **134T** of heat pipe **132T**, allowing the fan **280T** and heat sink **230T** to remove heat from said portions via the heat pipe **132T**.

(144) FIG. **18C** is a schematic view of a coupling mechanism **30A** for coupling the lid **L** and the vessel **120** for one or more implementations of the container system **100-100X** disclosed herein. In the illustrated embodiment, the lid **L** can be connected to one or more portions of the vessel **120** via a hinge that allows the lid **L** to be selectively moved between an open position (see FIG. **18C**) to allow access to the chamber **126**, and a closed position to disallow access to the chamber **126**.

(145) FIG. **18D** is a schematic view of another embodiment of a coupling mechanism **30B** between the lid **L** and the vessel **120** of the container system **100-100X**. In the illustrated embodiment, the lid **L** can have one or more electrical connectors **31B** that communicate with one or more electrical contacts **32B** on the vessel **120** when the lid **L** is coupled to the vessel **120**, thereby allowing operation of the fan **280**, TEC **220**, etc. that are optionally in the lid **L**. Optionally, one of the electrical connectors **31B** and electrical contacts **32B** can be contact pins (e.g., Pogo pins) and the other of the electrical connectors **31B** and electrical contacts **32B** can be electrical contact pads

(e.g., circular contacts) that optionally allows connection of the lid L to the vessel **120** irrespective of the angular orientation of the lid L relative to the vessel **120**.

(146) FIG. **18E** shows a schematic view of an embodiment of a vessel for the cooler container system, such as the cooler container systems **100-100X** disclosed herein. In the illustrated embodiment, the vessel **120** has electronics (e.g., one or more optional batteries, circuitry, optional transceiver) housed in a compartment E on a bottom of the vessel **120**. The electronics can communicate or connect to the fan **280**, TEC **220** or other components in the lid L via electrical connections (such as those shown and described in connection with FIG. **18D**), or via wires that extend through the hinge **30A** (such as that shown in FIG. **18C**).

(147) FIG. **18F** shows a schematic view of an embodiment of a vessel for the cooler container system, such as the cooler container systems **100-100X** disclosed herein. In the illustrated embodiment, the vessel **120** has electronics (e.g., one or more optional batteries, circuitry, optional transceiver) housed in a compartment E on a side of the vessel **120**. The electronics can communicate or connect to the fan **280**, TEC **220** or other components in the lid L via electrical connections (such as those shown and described in connection with FIG. **18D**), or via wires that extend through the hinge **30A** (such as that shown in FIG. **18C**).

(148) FIG. **19** shows another embodiment of a container system **100U** having a cooling system **200U**. The container system **100U** includes a vessel **120** with a chamber **126**. The vessel **120** can be double walled, as shown, with the space between the inner wall and outer wall under vacuum. A TEC **220U** can be in contact with a cold delivery member (e.g., stud) **225U**, which is in contact with the inner wall and can selectively thermally communicate with a hot side heat sink **230U**. The cold delivery member **225U** can be small relative to the size of the vessel **120**, and can extend through an opening **122U** in the vessel **120**. Optionally, the container system **100U** can have a pump P operable to pull a vacuum out from the cavity between the inner and outer walls of the vessel **120**.

(149) FIGS. **20-31** show a container system **100'** that includes a cooling system **200'**. The container system **100'** has a body **120'** that extends from a proximal end **122'** to a distal end **124'** and has an opening **123'** selectively closed by a lid L'. The body **120'** can optionally be box shaped. The lid L' can optionally be connected to the proximal end **122'** of the body **120'** by a hinge **130'** on one side of the body **120'**. A groove or handle **106'** can be defined on an opposite side of the body **120'** (e.g., at least partially defined by the lid L' and/or body **120'**), allowing a user to lift the lid L' to access a chamber **126'** in the container **100'**. Optionally, one or both of the lid L' and proximal end **122'** of the body **120'** can have one or more magnets (e.g., electromagnets, permanent magnets) that can apply a magnetic force between the lid L' and body **120'** to maintain the lid L' in a closed state over the body **120'** until a user overcomes said magnetic force to lift the lid L'. However, other suitable fasteners can be used to retain the lid L' in a closed position over the body **120'**.

(150) With reference to FIG. **27**, the body **120'** can include an outer wall **121'** and optionally include an inner wall **126A'** spaced apart from the outer wall **121'** to define a gap (e.g., annular gap, annular chamber) **128'** therebetween. Optionally, the inner wall **126A'** can be suspended relative to the outer wall **121'** in a way that provides the inner wall **126A'** with shock absorption (e.g., energy dissipation). For example, one or more springs can be disposed between the inner wall **126A'** and the outer wall **121'** that provide said shock absorption. Optionally, the container **100'** includes one or more accelerometers (e.g., in communication with the circuitry of the container **100'**) that sense motion (e.g., acceleration) of the container **100'**. Optionally, the one or more accelerometers communicate sensed motion information to the circuitry, and the circuitry optionally operates one or more components to adjust a shock absorption provided by the inner wall **126A'** (e.g., by tuning a shock absorption property of one or more springs, such as magnetorheological (MRE) springs) that support the inner surface **126A'**. In one implementation, the container **100'** can include a plastic and/or rubber structure in the gap **128'** between the inner wall **126A'** and the outer wall **121'** to aid in providing such shock absorption.

(151) The gap **128'** can optionally be filled with an insulative material (e.g., foam). In another implementation, the gap **128'** can be under vacuum. In still another implementation, the gap **128'** can be filled with a gas (e.g., air). Optionally, the inner wall **126A'** can be made of metal. Optionally, the outer wall **121'** can be made of plastic. In another implementation, the outer wall **121'** and the inner wall **126A'** are optionally made of the same material.

(152) With continued reference to FIG. 27, the cooling system **200'** can optionally be housed in a cavity **127'** disposed between a base **125'** of the container body **120'** and the inner wall **126A'**. The cooling system **200'** can optionally include one or more thermoelectric elements (TEC) (e.g., Peltier elements) **220'** in thermal communication with (e.g., in direct contact with) the inner wall **126A'**. In one implementation, the cooling system **200'** has only one TEC **220'**. The one or more TECs **220'** can optionally be in thermal communication with one or more heat sinks **230'**. Optionally, the one or more heat sinks **230'** can be a structure with a plurality of fins. Optionally, one or more fans **280'** can be in thermal communication with (e.g., in fluid communication with) the one or more heat sinks **230'**. The cooling system **200'** can optionally have one or more batteries **277'**, optionally have a converter **279'**, and optionally have a power button **290'**, that communicate with circuitry (e.g., on a printed circuit board **278'**) that controls the operation of the cooling system **200'**.

(153) The optional batteries **277'** provide power to one or more of the circuitry, one or more fans **280'**, one or more TECs **220'**, and one or more sensors (described further below). Optionally, at least a portion of the body **120'** (e.g., a portion of the base **125'**) of the container **100'** is removable to access the one or more optional batteries **277'**. Optionally, the one or more optional batteries **277'** can be provided in a removable battery pack, which can readily be removed and replaced from the container **100'**. Optionally, the container **100'** can include an integrated adaptor and/or retractable cable to allow connection of the container **100'** to a power source (e.g., wall outlet, vehicle power connector) to one or both of power the cooling system **200'** directly and charge the one or more optional batteries **277'**.

(154) With reference to FIGS. 22-23 and 27, the container system **100'** can have two or more handles **300** on opposite sides of the body **120'** to which a strap **400** can be removably coupled (see FIG. 24) to facilitate transportation of the container **100'**. For example, the user can carry the container **100'** by placing the strap **400** over their shoulder. Optionally, the strap **400** is adjustable in length. Optionally, the strap **400** can be used to secure the container system **100'** to a vehicle (e.g., moped, bicycle, motorcycle, etc.) for transportation. Optionally, the one or more handles **300** can be movable relative to the outer surface **121'** of the body **120'**. For example, the handles **300** can be selectively movable between a retracted position (see e.g., FIG. 22) and an extended position (see e.g., FIG. 23). Optionally, the handles **300** can be mounted within the body **120'** in a spring-loaded manner and be actuated in a push-to-open and push-to-close manner.

(155) With reference to FIGS. 26-27, the body **120'** can include one or more sets of vents on a surface thereof to allow air flow into and out of the body **120'**. For example, the body **120'** can have one or more vents **203'** defined on the bottom portion of the base **125'** of the body **120'** and can optionally have one or more vents **205'** at one or both ends of the base **125'**. Optionally, the vents **203'** can be air intake vents, and the vents **205'** can be air exhaust vents.

(156) With reference to FIG. 25A, the chamber **126** is optionally sized to receive and hold one or more trays **500** therein (e.g., hold a plurality of trays in a stacked configuration). Each tray **500** optionally has a plurality of receptacles **510**, where each receptacle **510** is sized to receive a container (e.g., a vial) **520** therein. The container **520** can optionally hold a liquid (e.g., a medication, such as insulin or a vaccine). Optionally, the tray **500** (e.g., the receptacle **510**) can releasably lock the containers **520** therein (e.g., lock the containers **520** in the receptacles **510**) to inhibit movement, dislodgement and/or damage to the containers **520** during transit of the container system **100'**. Optionally, the tray **500** can have one or more handles **530** to facilitate carrying of the tray **500** and/or pulling the tray **500** out of the chamber **126** or placing the tray **500** in the chamber

126. Optionally, the one or more handles **530** are movable between a retracted position (see FIG. **28**) and an extended position (see FIG. **26**). Optionally, the one or more handles **530** can be mounted within the tray **500** in a spring-loaded manner and be actuated in a push-to-extend and push-to-retract manner. In another implementation, the one or more handles **530** are fixed (e.g., not movable between a retracted and an extended position).

(157) With reference to FIGS. **25B-25D**, the tray **500** can include an outer tray **502** that removably receives one or more inner trays **504, 504'**, where different inner trays **504, 504'** can have a different number and/or arrangement of the plurality of receptacles **510** that receive the one or more containers (e.g., vials) **520** therein, thereby advantageously allowing the container **100'** to accommodate different number of containers **520** (e.g., for different medications, etc.). In one implementation, shown in FIG. **25C**, the inner tray **504** can have a relatively smaller number of receptacles **510** (e.g., sixteen), for example to accommodate relatively larger sized containers **520** (e.g., vials of medicine, such as vaccines and insulin, biological fluid, such as blood, etc.), and in another implementation, shown in FIG. **25D**, the inner tray **504'** can have a relatively larger number of receptacles **510** (e.g., thirty-eight), for example to accommodate relatively smaller sized containers **520** (e.g., vials of medicine, biological fluid, such as blood, etc.).

(158) With reference to FIG. **28**, the container system **100'** can have one or more lighting elements **550** that can advantageously facilitate users to readily see the contents in the chamber **126'** when in a dark environment (e.g., outdoors at night, in a rural or remote environment, such as mountainous, desert or rainforest region). In one implementation, the one or more lighting elements can be one or more light strips (e.g., LED strips) disposed at least partially on one or more surfaces of the chamber **126'** (e.g., embedded in a surface of the chamber **126'**, such as near the proximal opening of the chamber **126'**). Optionally, the one or more lighting elements **550** can automatically illuminate when the lid **L''** is opened. Once illuminated, the one or more lighting elements **550** can optionally automatically shut off when the lid **L''** is closed over the chamber **126'**. Optionally, the one or more lighting elements **550** can communicate with circuitry of the container **100'**, which can also communicate with a light sensor of the container **100'** (e.g., a light sensor disposed on an outer surface of the container **100'**). The light sensor can generate a signal when the sensed light is below a predetermined level (e.g., when container **100'** in a building without power or is in the dark, etc.) and communicate said signal to the circuitry, and the circuitry can operate the one or more lighting elements **550** upon receipt of such signal (e.g., and upon receipt of the signal indicating the lid **L''** is open).

(159) The container system **100'** can have a housing with one of a plurality of colors. Such different color housings can optionally be used with different types of contents (e.g., medicines, biological fluids), allowing a user to readily identify the contents of the container **100'** by its housing color. Optionally, such different colors can aid users in distinguishing different containers **100'** in their possession/use without having to open the containers **100'** to check their contents.

(160) With reference to FIGS. **29A-29C**, the container **100'** can optionally communicate (e.g., one-way communication, two-way communication) with one or more remote electronic device (e.g., mobile phone, tablet computer, desktop computer, remote server) **600**, via one or both of a wired or wireless connection (e.g., 802.11b, 802.11a, 802.11g, 802.11n standards, etc.). Optionally, the container **100'** can communicate with the remote electronic device **600** via an app (mobile application software) that is optionally downloaded (e.g., from the cloud) onto the remote electronic device **600**. The app can provide one or more graphical user interface screens **610A, 610B, 610C** via which the remote electronic device **600** can display one or more data received from the container **100'**. Optionally, a user can provide instructions to the container **100'** via one or more of the graphical user interface screens **610A, 610B, 610C** on the remote electronic device **600**.

(161) In one implementation, the graphical user interface (GUI) screen **610A** can provide one or more temperature presets corresponding to one or more particular medications (e.g., epinephrine/adrenaline for allergic reactions, insulin, vaccines, etc.). The GUI screen **610A** can

optionally allow the turning on and off of the cooling system **200'**. The GUI screen **610A** can optionally allow the setting of the control temperature to which the chamber **126'** in the container **100'** is cooled by the cooling system **200'**.

(162) In another implementation, the graphical user interface (GUI) screen **610B** can provide a dashboard display of one or more parameters of the container **100'** (e.g., ambient temperature, internal temperature in the chamber **126'**, temperature of the heat sink **230'**, temperature of the battery **277**, etc.). The GUI screen **610B** can optionally provide an indication (e.g., display) of power supply left in the one or more batteries **277** (e.g., % of life left, time remaining before battery power drains completely). Optionally, the GUI screen **610B** can also include information (e.g., a display) of how many of the receptacles **510** in the tray **500** are occupied (e.g., by containers **520**). Optionally, the GUI screen **610B** can also include information on the contents of the container **100'** (e.g., medication type or disease medication is meant to treat), information on the destination for the container **100'** and/or information (e.g., name, identification no.) for the individual assigned to the container **100'**.

(163) In another implementation, the GUI screen **610C** can include a list of notifications provided to the user of the container **100'**, including alerts on battery power available, alerts on ambient temperature effect on operation of container **100'**, alerts on a temperature of a heat sink of the container **100'**, alert on temperature of the chamber **126**, **126'**, **126V**, alert on low air flow through the intake vent **203'**, **203"**, **203V** and/or exhaust vent **205'**, **205"**, **205V** indicating they may be blocked/clogged, etc. One of skill in the art will recognize that the app can provide the plurality of GUI screens **610A**, **610B**, **610C** to the user, allowing the user to swipe between the different screens.

(164) Optionally, as discussed further below, the container **100'** can communicate information, such as temperature history of the chamber **126'** and/or first heat sink **210** that generally corresponds to a temperature of the containers **520**, **520V** (e.g., medicine containers, vials, cartridges, injectors), power level history of the batteries **277**, ambient temperature history, etc. to the cloud (e.g., on a periodic basis, such as every hour; on a continuous basis in real time, etc.) to one or more of a) an RFID tag on the container system **100**, **100'**, **100"**, **100B-100V** that can later be read (e.g., at the delivery location), b) to a remote electronic device (e.g., a mobile electronic device such as a smartphone or tablet computer or laptop computer or desktop computer), including wirelessly (e.g., via WiFi 802.11, BLUETOOTH®, or other RF communication), and c) to the cloud (e.g., to a cloud-based data storage system or server) including wirelessly (e.g., via WiFi 802.11, BLUETOOTH®, or other RF communication). Such communication can occur on a periodic basis (e.g., every hour; on a continuous basis in real time, etc.). Once stored on the RFID tag or remote electronic device or cloud, such information can be accessed via one or more remote electronic devices (e.g., via a dashboard on a smart phone, tablet computer, laptop computer, desktop computer, etc.). Additionally, or alternatively, the container system **100**, **100'**, **100"**, **100B-100V** can store in a memory (e.g., part of the electronics in the container system **100**, **100'**, **100"**, **100B-100V**) information, such as temperature history of the chamber **126**, **126'**, **126V**, temperature history of the first heat sink **210**, **210B-210V**, power level history of the batteries **277**, ambient temperature history, etc., which can be accessed from the container system **100**, **100'**, **100"**, **100B-100V** by the user via a wired or wireless connection (e.g., via the remote electronic device **600**).

(165) With reference to FIG. 30, the body **120'** of the container **100'** can have a visual display **140** on an outer surface **121'** of the body **120'**. The visual display **140'** can optionally display one or more of the temperature in the chamber **126'**, the ambient temperature, a charge level or percentage for the one or more batteries **277**, and amount of time left before recharging of the batteries **277** is needed. The visual display **140'** can include a user interface (e.g., pressure sensitive buttons, capacitance touch buttons, etc.) to adjust (up or down) the temperature preset at which the cooling system **200'** is to cool the chamber **126'** to. Accordingly, the operation of the container **100'** (e.g., of the cooling system **200'**) can be selected via the visual display and user interface **140'** on a surface

of the container **100'**. Optionally, the visual display **140'** can include one or more hidden-til-lit LEDs. Optionally, the visual display **140'** can include an electronic ink (e-ink) display. In one implementation, the container **100'** can optionally include a hidden-til-lit LED **142'** (see FIG. 34) that can selectively illuminate (e.g., to indicate one or more operating functions of the container **100'**, such as to indicate that the cooling system **200'** is in operation). The LED **142'** can optionally be a multi-color LED selectively operable to indicate one or more operating conditions of the container **100'** (e.g., green if normal operation, red if abnormal operation, such as low battery charge or inadequate cooling for sensed ambient temperature, etc.).

(166) With reference to FIG. 31, the container **100'** can include one or more security features that allow opening of the container **100'** only when the security feature(s) are met. In one implementation, the container **100'** can include a keypad **150** via which an access code can be entered to unlock the lid **L''** to allow access to the chamber **126'** when it matches the access code key programmed to the container **100'**. In another implementation, the container **100'** can additionally or alternatively have a biometric sensor **150'**, via which the user can provide a biometric identification (e.g., fingerprint) that will unlock the lid **L''** and allow access to the chamber **126'** when it matches the biometric key programmed to the container **100'**. Optionally, the container **100'** remains locked until it reaches its destination, at which point the access code and/or biometric identification can be utilized to unlock the container **100'** to access the contents (e.g., medication) in the chamber **126'**.

(167) The container **100'** can optionally be powered in a variety of ways. In one implementation, the container system **100'** is powered using 12 VDC power (e.g., from one or more batteries **277'**). In another implementation, the container system **100'** is powered using 120 VAC or 240 VAC power. In another implementation, the cooling system **200'** can be powered via solar power. For example, the container **100'** can be removably connected to one or more solar panels so that electricity generated by the solar panels is transferred to the container **100'**, where circuitry of the container **100'** optionally charges the one or more batteries **277** with the solar power. In another implementation, the solar power from said one or more solar panels directly operates the cooling system **200'** (e.g., where batteries **277** are excluded from the container **100'**). The circuitry in the container **100'** can include a surge protector to inhibit damage to the electronics in the container **100'** from a power surge.

(168) In operation, the cooling system **200'** can optionally be actuated by pressing the power button **290**. Optionally, the cooling system **200'** can additionally (or alternatively) be actuated remotely (e.g., wirelessly) via a remote electronic device, such as a mobile phone, tablet computer, laptop computer, etc. that wirelessly communicates with the cooling system **200'** (e.g., with a receiver or transceiver of the circuitry). The chamber **126'** can be cooled to a predetermined and/or a user selected temperature or temperature range. The user selected temperature or temperature range can be selected via a user interface on the container **100'** and/or via the remote electronic device.

(169) The circuitry optionally operates the one or more TECs **220'** so that the side of the one or more TECs **220'** adjacent the inner wall **126A'** is cooled and so that the side of the one or more TECs **220'** adjacent the one or more heat sinks **230'** is heated. The TECs **220'** thereby cool the inner wall **126A'** and thereby cools the chamber **126'** and the contents (e.g., tray **500** with containers (e.g., vials) **520** therein). Though not shown in the drawings, one or more sensors (e.g., temperature sensors) are in thermal communication with the inner wall **126A'** and/or the chamber **126'** and communicate information to the circuitry indicative of the sensed temperature. The circuitry operates one or more of the TECs **220'** and one or more fans **280'** based at least in part on the sensed temperature information to cool the chamber **126'** to the predetermined temperature and/or user selected temperature. The circuitry operates the one or more fans **280'** to flow air (e.g., received via the intake vents **203'**) over the one or more heat sinks **230'** to dissipate heat therefrom, thereby allowing the one or more heat sinks **230'** to draw more heat from the one or more TECs **220'**, which in turn allows the one or more TECs **220'** to draw more heat from (i.e., cool) the inner

wall **126A'** to thereby further cool the chamber **126'**. Said air flow, once it passes over the one or more heat sinks **230'**, is exhausted from the body **120'** via the exhaust vents **205'**.

(170) FIGS. **32-34** schematically illustrate a container **100''** that includes a cooling system **200''**. The container system **100''** can include a vessel body **120** removably sealed by a lid **L'''**. Some of the features of the container **100''** and cooling system **200''** are similar to the features of the container **100'** and cooling system **200'** in FIGS. **20-31**. Thus, reference numerals used to designate the various components of the container **100''** and cooling system **200''** are similar to those used for identifying the corresponding components of the cooling system **200'** in FIGS. **20-31**, except that an "''" is used. Therefore, the structure and description for said components of the cooling system **200'** of FIGS. **20-31** are understood to also apply to the corresponding components of the container **100''** and cooling system **200''** in FIGS. **32-34**, except as described below. FIG. **33A** is a front view of the container **100''** in FIG. **32**. FIG. **33B** is a smaller version of the container **100''** and optionally has the same internal components as shown for the container in FIG. **33A** (e.g., as shown in FIGS. **37-39**).

(171) With reference to FIGS. **32-34**, the container **100''** differs from the container **100'** in that the container **100''** has a generally cylindrical or tube-like body **120''** with a generally cylindrical outer surface **121''**. The container **100''** can have similar internal components as the container **100'**, such as a chamber **126''** defined by an inner wall **126A''**, TEC **220''**, heat sink **230''**, one or more fans **280''**, one or more optional batteries **277''**, converter **279''** and power button **290''**. The lid **L'''** can have one or more vents **203''**, **205''** defined therein, and operate in a similar manner as the vents **203'**, **205'** described above. The container **100''** can have a variety of sizes (see FIGS. **33A-33B**) that can accommodate a different number and/or size of containers **520''**. The container **100''** and cooling system **200''** operate in a similar manner described above for the container **100'** and cooling system **200'**.

(172) The container **100''** can optionally include a display similar to the display **140'** described above for the container **100'** (e.g., that displays one or more of the temperature in the chamber **126''**, the ambient temperature, a charge level or percentage for the one or more batteries **277''**, and amount of time left before recharging of the batteries **277''** is needed). The container **100''** can optionally include a hidden-til-lit LED **142''** (see FIG. **36**) that can selectively illuminate (e.g., to indicate one or more operating functions of the container **100''**, such as to indicate that the cooling system **200'** is in operation). The LED **142''** can optionally be a multi-color LED selectively operable to indicate one or more operating conditions of the container **100''** (e.g., green if normal operation, red if abnormal operation, such as low battery charge or inadequate cooling for sensed ambient temperature, etc.).

(173) With reference to FIG. **34**, the container **100''** can be removably placed on a base **700''**, which can connect to a power source (e.g., wall outlet) via a cable **702''**. In one implementation, the base **700''** directly powers the cooling system **200''** of the container **100''** (e.g., to cool the contents in the container **100''**) to the desired temperature (e.g., the temperature required by the medication, such as insulin, in the chamber **126''** of the container **100''**). In another implementation, the base **700''** can additionally or alternatively charge the one or more optional batteries **277''**, so that the batteries **277''** take over powering of the cooling system **200''** when the container **100''** is removed from the base **700''**. Optionally, the vessel **120''** of the container system **100''** can have one or more electrical contacts **EC1** (e.g., contact rings) that communicate with one or more electrical contacts **EC2** (e.g., pogo pins) of the base **700''** when the vessel **120''** is placed on the base **700''**. In another implementation, the base **700''** can transfer power to the vessel **120''** of the container system **100''** via inductive coupling (e.g., electromagnetic induction).

(174) With reference to FIGS. **35A-35C**, the container **100''** can optionally communicate (e.g., one-way communication, two-way communication) with one or more remote electronic device (e.g., mobile phone, tablet computer, desktop computer) **600**, via one or both of a wired or wireless connection. Optionally, the container **100''** can communicate with the remote electronic device **600**

via an app (mobile application software) that is optionally downloaded (e.g., from the cloud) onto the remote electronic device **600**. The app can provide one or more graphical user interface screens **610A"**, **610B"**, **610C"** via which the remote electronic device **600** can display one or more data received from the container **100"**. Optionally, a user can provide instructions to the container **100"** via one or more of the graphical user interface screens **610A"**, **610B"**, **610C"** on the remote electronic device **600**.

(175) In one implementation, the graphical user interface (GUI) screen **610A"** can provide one or more temperature presets corresponding to one or more particular medications (e.g., insulin). The GUI **610A"** can optionally allow the turning on and off of the cooling system **200"**. The GUI **610A"** can optionally allow the setting of the control temperature to which the chamber **126"** in the container **100"** is cooled by the cooling system **200"**.

(176) In another implementation, the graphical user interface (GUI) screen **610B"** can provide a dashboard display of one or more parameters of the container **100"** (e.g., ambient temperature, internal temperature in the chamber **126"**, etc.). The GUI screen **610B"** can optionally provide an indication (e.g., display) of power supply left in the one or more batteries **277"** (e.g., % of life left, time remaining before battery power drains completely). Optionally, the GUI screen **610B"** can also include information (e.g., a display) of how many of the receptacles **510"** in the tray **500"** are occupied (e.g., by containers **520"**). Optionally, the GUI screen **610B"** can also include information on the contents of the container **100'** (e.g., medication type or disease medication is meant to treat), information on the physician (e.g., name of doctor and contact phone no) and/or information (e.g., name, date of birth, medical record no.) for the individual assigned to the container **100"**.

(177) In another implementation, the GUI screen **610C"** can include a list of notifications provided to the user of the container **100"**, including alerts on battery power available, alerts on ambient temperature effect on operation of container **100"**, etc. One of skill in the art will recognize that the app can provide the plurality of GUI screens **610A"**, **610B"**, **610C"** to the user, allowing the user to swipe between the different screens. Optionally, as discussed further below, the container **100"** can communicate information, such as temperature history of the chamber **126"**, power level history of the batteries **277"**, ambient temperature history, etc. to the cloud (e.g., on a periodic basis, such as every hour; on a continuous basis in real time, etc.).

(178) In some implementations, the container system **100**, **100'**, **100"**, **100B-100X** can include one or both of a radiofrequency identification (RFID) reader and a barcode reader. For example, the RFID reader and/or barcode reader can be disposed proximate (e.g., around) a rim of the chamber **126**, **126'**, **126"** to that it can read content units (e.g., vials, containers) placed into or removed from the chamber **126**, **126'**, **126"**. The RFID reader or barcode reader can communicate data to the circuitry in the container system, which as discussed above, can optionally store such data in a memory or the container system and/or communicate such data to a separate or remote computing system, such as a remote computer server (e.g., accessible by a doctor treating the patient with the medication in the container), a mobile electronic device, such as a mobile phone or tablet computer. Such communication can optionally be in one or both of a wired manner (via a connector on the container body) or wireless manner (via a transmitter or transceiver of the container in communication with the circuitry of the container). Each of the contents placed in the chamber of the container (e.g., each medicine unit, such as each vial or container) optionally has an RFID tag or barcode that is read by the RFID reader or barcode reader as it is placed in and/or removed from the chamber of the container, thereby allowing the tracking of the contents of the container system **100**, **100'**, **100"**, **100B-100X**. Optionally, the container system (e.g., the RFID reader, barcode reader and/or circuitry) of the container system, send a notification (e.g., to a remote computer server, to one or more computing systems, to a mobile electronic device such as a smartphone or tablet computer or laptop computer or desktop computer) every time a medicine unit (e.g., vial, container) is placed into and/or removed from the chamber of the container system **100**, **100'**, **100"**, **100B-100X**.

(179) In some implementations, the container system **100, 100', 100'', 100B-100X** can additionally or alternatively (to the RFID reader and/or barcode reader) include a proximity sensor, for example in the chamber **126, 126', 126''** to advantageously track one or both of the insertion of and removal of content units (e.g., medicine units such as vials, containers, pills, etc.) from the container system. Such a proximity sensor can communicate with the circuitry of the container and advantageously facilitate tracking, for example, of the user taking medication in the container, or the frequency with which the user takes the medication. Optionally, operation of the proximity sensor can be triggered by a signal indicating the lid **L, L', L''** has been opened. The proximity sensor can communicate data to the circuitry in the container system, which as discussed above, can optionally store such data in a memory or the container system and/or communicate such data to a separate or remote computing system, such as a remote computer server (e.g., accessible by a doctor treating the patient with the medication in the container), a mobile electronic device, such as a mobile phone or tablet computer. Such communication can optionally be in one or both of a wired manner (via a connector on the container body) or wireless manner (via a transmitter or transceiver of the container in communication with the circuitry of the container).

(180) In some implementations, the container system **100, 100', 100'', 100B-100X** can additionally or alternatively (to the RFID reader and/or barcode reader) include a weight sensor, for example in the chamber **126, 126', 126''** to advantageously track the removal of content units (e.g. medicine units such as vials, containers, pills, etc.) from the container system. Such a weight sensor can communicate with the circuitry of the container and advantageously facilitate tracking, for example, of the user taking medication in the container, or the frequency with which the user takes the medication. Optionally, operation of the weight sensor can be triggered by a signal indicating the lid **L, L', L''** has been opened. The weight sensor can communicate data to the circuitry in the container system, which as discussed above, can optionally store such data in a memory or the container system and/or communicate such data to a separate or remote computing system, such as a remote computer server (e.g., accessible by a doctor treating the patient with the medication in the container), a mobile electronic device, such as a mobile phone or tablet computer. Such communication can optionally be in one or both of a wired manner (via a connector on the container body) or wireless manner (via a transmitter or transceiver of the container in communication with the circuitry of the container).

(181) FIG. **36** shows a container system, such as the container systems **100, 100', 100'', 100A-100X** described herein, removably connectable to a battery pack **B** (e.g., a Dewalt battery pack), which can provide power to one or more electrical components (e.g., TEC, fan, circuitry, etc.) of the container systems or the cooling systems **200, 200', 200'', 200A-200T**. Optionally, the vessel **120** of the container system can have one or more electrical contacts **EC1** (e.g., contact rings) that communicate with one or more electrical contacts **EC2** (e.g., pogo pins) when the vessel **120** is placed on the battery pack **B**. In another implementation, the battery pack **B** can transfer power to the vessel **120** of the container system via inductive coupling (e.g., electromagnetic induction).

(182) FIGS. **37-39** show a schematic cross-sectional view of a container system **100V** that includes a cooling system **200V**. Optionally, the container system **100V** has a container vessel **120V** that is optionally cylindrical and symmetrical about a longitudinal axis, and one of ordinary skill in the art will recognize that at least some of the features shown in cross-section in FIGS. **37-39** are defined by rotating them about the axis to define the features of the container **100V** and cooling system **200V**. Some of the features of the cooling system **200V**, which optionally serves as part of the lid **L'** that selectively seals the vessel **120V**, are similar to features in the cooling system **200M** in FIGS. **13A-13B**. Thus, references numerals used to designate the various components of the cooling system **200V** are similar to those used for identifying the corresponding components of the cooling system **200M** in FIGS. **13A-13B**, except that an “V” is used. Therefore, the structure and description for said similar components of the cooling system **200M** in FIGS. **13A-13B** are understood to also apply to the corresponding components of the cooling system **200V** in FIGS. **37-**

39, except as described below.

(183) With reference to FIGS. 37-39, the cooling system 200V can include a heat sink (cold side heat sink) 210V in thermal communication with a thermoelectric element (TEC) 220V and can be in thermal communication with the chamber 126V of the vessel 120V. Optionally, the cooling system 200V can include a fan 216V selectively operable to draw air from the chamber 126V into contact with the cold side heat sink 210V. Optionally, cooling system 200V can include an insulator member 270V disposed between the heat sink 210V and an optional lid top plate 202V, where the lid top plate 202V is disposed between the heat sink (hot side heat sink) 230V and the insulator 270V, the insulator 270V disposed about the TEC 220V. As shown in FIG. 39, air flow Fr is drawn by the fan 216V from the chamber 126V and into contact with the heat sink (cold side heat sink) 210V (e.g., to cool the air flow Fr), and then returned to the chamber 126V. Optionally, the air flow Fr is returned via one or more openings 218V in a cover plate 217V located distally of the heat sink 210V and fan 216V.

(184) With continued reference to FIGS. 37-39, the TEC 220V is selectively operated to draw heat from the heat sink (e.g., cold-side heat sink) 210V and transfer it to the heat sink (hot-side heat sink) 230V. A fan 280V is selectively operable to dissipate heat from the heat sink 230V, thereby allowing the TEC 220V to draw further heat from the chamber 126V via the heat sink 210V. As shown in FIG. 40, during operation of the fan 280V, intake air flow Fi is drawn through one or more openings 203V in the lid cover L''' and over the heat sink 230V (where the air flow removes heat from the heat sink 230V), after which the exhaust air flow Fe flows out of one or more openings 205V in the lid cover L'''. Optionally, both the fan 280V and the fan 216V are operated simultaneously. In another implementation, the fan 280V and the fan 216V are operated at different times (e.g., so that operation of the fan 216V does not overlap with operation of the fan 280V).

(185) As shown in FIGS. 37-39, the chamber 126V optionally receives and holds one or more (e.g., a plurality of) trays 500V, each tray 500V supporting one or more (e.g., a plurality of) liquid containers 520V (e.g., vials, such as vaccines, medications, etc.). The lid L''' can have a handle 400V used to remove the lid L''' from the vessel 120V to remove contents from the chamber 126V or place contents in the chamber 126V (e.g., remove the trays 500 via handle 530V). The lid L''' can have a scaling gasket G, such as disposed circumferentially about the insulator 270V to seal the lid L''' against the chamber 126V. The inner wall 136V of the vessel 120V is spaced from the outer wall 121V to define a gap (e.g., an annular gap) 128V therebetween. Optionally, the gap 128V can be under vacuum. Optionally, the inner wall 136V defines at least a portion of an inner vessel 130V. Optionally, the inner vessel 130V is disposed on a bottom plate 272V.

(186) The bottom plate 272V can be spaced from a bottom 275V of the vessel 120V to define a cavity 127V therebetween. The cavity 127V can optionally house one or more batteries 277V, a printed circuit board (PCBA) 278V and at least partially house a power button or switch 290V. Optionally, the bottom 275V defines at least a portion of an end cap 279V attached to the outer wall 121V. Optionally, the end cap 279V is removable to access the electronics in the cavity 127V (e.g., to replace the one or more batteries 277V, perform maintenance on the electronics, such as the PCBA 278V, etc.). The power button or switch 290V is accessible by a user (e.g., can be pressed to turn on the cooling system 200V, pressed to turn off the cooling system 200V, pressed to pair the cooling system 200V with a mobile electronic device, etc.). As shown in FIG. 37, the power switch 290V can be located generally at the center of the end cap 279V (e.g., so that it aligns/extends along the longitudinal axis of the vessel 120V).

(187) The electronics (e.g., PCBA 278V, batteries 277V) can electrically communicate with the fans 280V, 216V and TEC 220V in the lid L'' via one or more electrical contacts (e.g., electrical contact pads, Pogo pins) in the lid L'' that contact one or more electrical contacts (e.g., Pogo pins, electrical contact pads) in the portion of the vessel 120V that engages the lid L''', such as in a similar manner to that described above for FIG. 18D.

(188) FIG. 40 shows a block diagram of a communication system for (e.g., incorporated into) the

devices described herein (e.g., the one or more container systems **100**, **100'**, **100''**, **100A-100X**). In the illustrated embodiment, circuitry EM can receive sensed information from one or more sensors S1-Sn (e.g., level sensors, volume sensors, temperature sensors, battery charge sensors, biometric sensors, load sensors, Global Positioning System or GPS sensors, radiofrequency identification or RFID reader, etc.). The circuitry EM can be housed in the container, such as in the vessel **120** (e.g., bottom of vessel **120**, side of vessel **120**, as discussed above) or in a lid L of the container. The circuitry EM can receive information from and/or transmit information (e.g., instructions) to one or more heating or cooling elements HC, such as the TEC **220**, **220'**, **220A-220X** (e.g., to operate each of the heating or cooling elements in a heating mode and/or in a cooling mode, turn off, turn on, vary power output of, etc.) and optionally to one or more power storage devices PS (e.g., batteries, such as to charge the batteries or manage the power provided by the batteries to the one or more heating or cooling elements).

(189) Optionally, the circuitry EM can include a wireless transmitter, receiver and/or transceiver to communicate with (e.g., transmit information, such as sensed temperature and/or position data, to and receive information, such as user instructions, from one or more of: a) a user interface UI1 on the unit (e.g., on the body of the vessel **120**), b) an electronic device ED (e.g., a mobile electronic device such as a mobile phone, PDA, tablet computer, laptop computer, electronic watch, a desktop computer, remote server), c) via the cloud CL, or d) via a wireless communication system such as WiFi and/or Bluetooth BT. The electronic device ED can have a user interface UI2, that can display information associated with the operation of the container system (such as the interfaces disclosed above, see FIGS. **29A-29C**, **35A-35C**), and that can receive information (e.g., instructions) from a user and communicate said information to the container system **100**, **100'**, **100''**, **100A-100X** (e.g., to adjust an operation of the cooling system **200**, **200'**, **200''**, **200A-200X**).

(190) In operation, the container system can operate to maintain the chamber **126** of the vessel **120** at a preselected temperature or a user selected temperature. The cooling system can operate the one or more TECs to cool the chamber **126** (e.g., if the temperature of the chamber is above the preselected temperature, such as when the ambient temperature is above the preselected temperature) or to heat the chamber **126** (e.g., if the temperature of the chamber **126** is below the preselected temperature, such as when the ambient temperature is below the preselected temperature). The preselected temperature may be tailored to the contents of the container (e.g., a specific medication, a specific vaccine), and can be stored in a memory of the container, and the cooling system or heating system, depending on how the temperature control system is operated, can operate the TEC to approach the preselected or set point temperature.

(191) Optionally, the circuitry EM can communicate (e.g., wirelessly) information to a remote location (e.g., cloud-based data storage system, remote computer, remote server, mobile electronic device such as a smartphone or tablet computer or laptop or desktop computer) and/or to the individual carrying the container (e.g., via their mobile phone, via a visual interface on the container, etc.), such as a temperature history of the chamber **126** to provide a record that can be used to evaluate the efficacy of the medication in the container and/or alerts on the status of the medication in the container. Optionally, the temperature control system (e.g., cooling system, heating system) automatically operates the TEC to heat or cool the chamber **126** of the vessel **120** to approach the preselected temperature. In one implementation, the cooling system **200**, **200'**, **200''**, **200B-200X** can cool and maintain one or both of the chamber **126**, **126'**, **126V** and the containers **520**, **520V** at or below 15 degrees Celsius, such as at or below 10 degrees Celsius, in some examples at approximately 5 degrees Celsius.

(192) In one implementation, the one or more sensors S1-Sn can include one more air flow sensors in the lid L that can monitor airflow through one or both of the intake vent **203'**, **203''**, **203V** and exhaust vent **205'**, **205''**, **205V**. If said one or more flow sensors senses that the intake vent **203'**, **203''**, **203V** is becoming clogged (e.g., with dust) due to a decrease in air flow, the circuitry EM (e.g., on the PCBA **278V**) can optionally reverse the operation of the fan **280**, **280'**, **280B-280P**,

280V for one or more predetermined periods of time to draw air through the exhaust vent **205'**, **205''**, **205V** and exhaust air through the intake vent **203'**, **203''**, **203V** to clear (e.g., unclog, remove the dust from) the intake vent **203'**, **203''**, **203V**. In another implementation, the circuitry EM can additionally or alternatively send an alert to the user (e.g., via a user interface on the container **100**, **100'**, **100''**, **100B-100X**, wirelessly to a remote electronic device such as the user's mobile phone via GUI **610A-610C**, **610A'-610C'**) to inform the user of the potential clogging of the intake vent **203'**, **203''**, **203V**, so that the user can inspect the container **100**, **100'**, **100''**, **100B-100X** and can instruct the circuitry EM (e.g., via an app on the user's mobile phone) to run a "cleaning" operation, for example, by running the fan **280**, **280'**, **280B-280P**, **280V** in reverse to exhaust air through the intake vent **203'**, **203''**, **203V**.

(193) In one implementation, the one or more sensors **S1-Sn** can include one more Global Positioning System (GPS) sensors for tracking the location of the container system **100**, **100'**, **100''**, **100B-100X**. The location information can be communicated, as discussed above, by a transmitter and/or transceiver associated with the circuitry EM to a remote location (e.g., a mobile electronic device, a cloud-based data storage system, etc.).

(194) FIG. **41A** shows a container system **100X** (e.g., a medicine cooler container) that includes a cooling system **200X**. Though the container system **100X** has a generally box shape, in other implementations it can have a generally cylindrical or tube shape, similar to the container system **100**, **100''**, **100B**, **100C**, **100D**, **100E**, **100F**, **100G**, **100H**, **100I**, **100J**, **100K**, **100K'**, **100L**, **100L'**, **100M**, **100N**, **100P**, **100Q**, **100R**, **100T**, **100U**, **100V**, or the features disclosed below for container system **100X** can be incorporated into the generally cylindrical or tube shaped containers noted above. In other implementations, the features disclosed below for container system **100X** can be incorporated into containers **100'** disclosed above. In one implementation, the cooling system **200X** can be in the lid **L** of the container system **100X** and can be similar to (e.g., have the same or similar components as) the cooling system **200**, **200''**, **200B**, **200B'**, **200C**, **200D**, **200E**, **200F**, **200G**, **200H**, **200I**, **200J**, **200K**, **200K'**, **200L**, **200L'**, **200M**, **200N**, **200P**, **200Q**, **200R**, **200S**, **200T**, **200V** described above. In another implementation, the cooling system can be disposed in a portion of the container vessel **120X** (e.g. a bottom portion of the container vessel **120X**, similar to cooling system **200'** in vessel **120'** described above).

(195) As shown in FIG. **41A**, the container system **100X** can include a display screen **188X**. Though FIG. **41A** shows the display screen **188X** on the lid **L**, it can alternatively (or additionally) be incorporated into a side surface **122X** of the container vessel **120X**. The display screen **188X** can optionally be an electronic ink or E-ink display (e.g., electrophoretic ink display). In another implementation, the display screen **188X** can be a digital display (e.g., liquid crystal display or LCD, light emitting diode or LED, etc.). Optionally, the display screen **188X** can display a label **189X** (e.g., a shipping label with one or more of an address of sender, an address of recipient, a Maxi Code machine readable symbol, a QR code, a routing code, a barcode, and a tracking number), but can optionally additionally or alternatively display other information (e.g., temperature history information, information on the contents of the container system **100X**). The container system **100X** can optionally also include a user interface **184X**. In FIG. **41A**, the user interface **184X** is a button on the lid **L**. In another implementation, the user interface **184X** is disposed on the side surface **122X** of the container vessel **120X**. In one implementation, the user interface **184X** is a depressible button. In another implementation, the user interface **184X** is a capacitive sensor (e.g., touch sensitive sensor). In another implementation, the user interface **184X** is a sliding switch (e.g., sliding lever). In another implementation, the user interface **184X** is a rotatable dial. In still another implementation, the user interface **184X** can be a touch screen portion (e.g., separate from or incorporated as part of the display screen **188X**). Advantageously, actuation of the user interface **184X** can alter the information shown on the display **188X**, such as the form of a shipping label shown on an E-ink display **188X**. For example, actuation of the user interface **184X**, can switch the text associated with the sender and receiver, allowing the container system

100X to be shipped back to the sender once the receiving party is done with it.

(196) FIG. **41B** shows a block diagram of electronics **180** of the container system **100X**. The electronics **180** can include circuitry EM' (e.g., including one or more processors on a printed circuit board). The circuitry EM' communicate with one or more batteries PS', with the display screen **188X**, and with the user interface **184X**. Optionally, a memory module **185X** is in communication with the circuitry EM'. In one implementation, the memory module **185X** can optionally be disposed on the same printed circuit board as other components of the circuitry EM'. The circuitry EM' optionally controls the information displayed on the display screen **188X**. Information (e.g., sender address, recipient address, etc.) can be communicated to the circuitry EM' via an input module **186X**. The input module **186X** can receive such information wirelessly (e.g., via radiofrequency or RF communication, via infrared or IR communication, via WiFi 802.11, via BLUETOOTH®, etc.), such as using a wand (e.g., a radiofrequency or RF wand that is waved over the container system **100X**, such as over the display screen **188X**, where the wand is connected to a computer system where the shipping information is contained). Once received by the input module **186X**, the information (e.g., shipping information for a shipping label to be displayed on the display screen **188X** can be electronically saved in the memory module **185X**). Advantageously, the one or more batteries PS' can power the electronics **180**, and therefore the display screen **188X** for a plurality of uses of the container **100X** (e.g., during shipping of the container system **100X** up to one-thousand times).

(197) FIG. **42A** shows a block diagram of one method **800A** for shipping the container system **100X**. At step **810**, one or more containers, such as containers **520** (e.g., medicine containers, such as vials, cartridges (such as for injector pens), injector pens, vaccines, medicine such as insulin, epinephrine, etc.) are placed in the container vessel **120X** of the container system **100X**, such as at a distribution facility for the containers **520**. At step **820**, the lid L is closed over the container vessel **120X** once finished loading all containers **520** into the container vessel **120X**. Optionally, the lid L is locked to the container vessel **120X** (e.g., via a magnetically actuated lock, including an electromagnet actuated when the lid is closed that can be turned off with a code, such as a digital code). At step **830**, information (e.g., shipping label information) is communicated to the container system **100X**. For example, as discussed above, a radiofrequency (RF) wand can be waved over the container system **100X** (e.g., over the lid L) to transfer the shipping information to the input module **186X** of the electronics **80** of the container system **100X**. At step **840**, the container system **100X** is shipped to the recipient (e.g., displayed on the shipping label **189X** on the display screen **188X**).

(198) FIG. **42B** shows a block diagram of a method **800B** for returning the container **100X**. At step **850**, after receiving the container system **100X**, the lid L can be opened relative to the container vessel **120X**. Optionally, prior to opening the lid L, the lid L is unlocked relative to the container vessel **100X** (e.g., using a code, such as a digital code, provided to the recipient from the shipper) via keypad and/or biometric identification (e.g., fingerprint on the container vessel, as discussed above with respect to FIG. **31**). At step **860**, the one or more containers **520** are removed from the container vessel **120X**. At step **870**, the lid L is closed over the container vessel **120X**. At step **880**, the user interface **184X** (e.g., button) is actuated to switch the information of the sender and recipient in the display screen **188X** with each other, advantageously allowing the return of the container system **100X** to the original sender to be used again without having to reenter shipping information on the display screen **188X**. The display screen **188X** and label **189X** advantageously facilitate the shipping of the container system **100X** without having to print any separate labels for the container system **100X**. Further, the display screen **188X** and user interface **184X** advantageously facilitate return of the container system **100X** to the sender (e.g. without having to reenter shipping information, without having to print any labels), where the container system **100X** can be reused to ship containers **520** (e.g., medicine containers, such as vials, cartridges (such as for injector pens), injector pens, vaccines, medicine such as insulin, epinephrine, etc.) again, such

as to the same or a different recipient. The reuse of the container system **100K** for delivery of perishable material (e.g., medicine) advantageously reduces the cost of shipping by allowing the reuse of the container vessel **120X** (e.g., as compared to commonly used cardboard containers, which are disposed of after one use).

ADDITIONAL EMBODIMENTS

(199) In embodiments of the present invention, a portable cooler container with active temperature control, may be in accordance with any of the following clauses:

(200) Clause 1. A portable cooler container with active temperature control, comprising: a container body having a chamber configured to receive and hold one or more containers of medicine; a lid removably coupleable to the container body to access the chamber; and a temperature control system comprising one or more thermoelectric elements configured to actively heat or cool at least a portion of the chamber, one or more batteries, circuitry configured to control an operation of the one or more thermoelectric elements to heat or cool at least a portion of the chamber to a predetermined temperature or temperature range; and a display screen disposed on one or both of the container body and the lid, the display screen configured to selectively display shipping information for the portable cooler container using electronic ink.

(201) Clause 2. The portable cooler container of any preceding clause, further comprising a button or touch screen actuable by a user to automatically switch sender and recipient information on the display screen to facilitate return of the portable cooler container to a sender.

(202) Clause 3. The portable cooler container of any preceding clause, wherein the body comprises an outer peripheral wall and a bottom portion attached to the outer peripheral wall, the inner peripheral wall being spaced relative to the outer peripheral wall to define a gap between the inner peripheral wall and the outer peripheral wall, the base spaced apart from the bottom portion to define a cavity between the base and the bottom portion, the one or more batteries and circuitry at least partially disposed in the cavity.

(203) Clause 4. The portable cooler container of any preceding clause, wherein the one or more thermoelectric elements are housed in the lid, the temperature control system further comprising a first heat sink unit in thermal communication with one side of the one or more thermoelectric elements, a second heat sink unit in thermal communication with an opposite side of the one or more thermoelectric elements, and one or more fans, wherein the one or more fans, first heat sink unit and second heat sink unit are at least partially housed in the lid, the first heat sink configured to heat or cool at least a portion of the chamber.

(204) Clause 5. The portable cooler container of any preceding clause, further comprising one or more sensors configured to sense the one or more parameters of the chamber or temperature control system and to communicate the sensed information to the circuitry.

(205) Clause 6. The portable cooler container of any preceding clause, wherein at least one of the one or more sensors is a temperature sensor configured to sense a temperature in the chamber and to communicate the sensed temperature to the circuitry, the circuitry configured to communicate the sensed temperature data to the cloud-based data storage system or remote electronic device.

(206) Clause 7. The portable cooler container of any preceding clause, further comprising one or more electrical contacts on a rim of the container body configured to contact one or more electrical contacts on the lid when the lid is coupled to the container body so that the circuitry controls the operation of the one or more thermoelectric elements and one or more fans when the lid is coupled to the container body.

(207) Clause 8. The portable cooler container of any preceding clause, wherein the gap is under vacuum.

(208) Clause 9. The portable cooler container of any preceding clause, further comprising a removable tray configured to removably receive the containers of medicine therein and to releasably lock the containers in the tray to inhibit dislodgement of the medicine containers from the tray during shipping of the portable cooler container.

(209) Clause 10. The portable cooler container of any preceding clause, further comprising means for thermally disconnecting the one or more thermoelectric elements from the chamber to inhibit heat transfer between the one or more thermoelectric elements and the chamber.

(210) Clause 11. A portable cooler container with active temperature control, comprising: a container body having a chamber configured to receive and hold one or more medicine containers, the chamber defined by a base and an inner peripheral wall of the container body; a lid removably coupleable to the container body to access the chamber; and a temperature control system comprising one or more thermoelectric elements and one or more fans, one or both of the thermoelectric elements and fans configured to actively heat or cool at least a portion of the chamber, one or more batteries, and circuitry configured to control an operation of the one or more thermoelectric elements to heat or cool at least a portion of the chamber to a predetermined temperature or temperature range.

(211) Clause 12. The portable cooler container of clause 11, wherein the body comprises an outer peripheral wall and a bottom portion attached to the outer peripheral wall, the inner peripheral wall being spaced relative to the outer peripheral wall to define a gap between the inner peripheral wall and the outer peripheral wall, the base spaced apart from the bottom portion to define a cavity between the base and the bottom portion, the one or more batteries and circuitry at least partially disposed in the cavity.

(212) Clause 13. The portable cooler container of any of clauses 11-12, wherein the one or more thermoelectric elements are housed in the lid, the temperature control system further comprising a first heat sink unit in thermal communication with one side of the one or more thermoelectric elements, a second heat sink unit in thermal communication with an opposite side of the one or more thermoelectric elements, wherein the one or more fans, first heat sink unit and second heat sink unit are at least partially housed in the lid, the first heat sink configured to heat or cool at least a portion of the chamber.

(213) Clause 14. The portable cooler container of any of clauses 11-13, further comprising one or more sensors, at least one of the one or more sensors is a temperature sensor configured to sense a temperature in the chamber and to communicate the sensed temperature to the circuitry.

(214) Clause 15. The portable cooler container of any of clauses 11-14, wherein the circuitry further comprises a transmitter configured to transmit one or both of temperature and position information for the portable cooler container to one or more of a memory of the portable cooler container, a radiofrequency identification tag of the portable cooler containers, a cloud-based data storage system, and a remote electronic device.

(215) Clause 16. The portable cooler container of any of clauses 11-15, further comprising a display on one or both of the container body and the lid, the display configured to display information indicative of a temperature of the chamber.

(216) Clause 17. The portable cooler container of any of clauses 11-16, further comprising one or more electrical contacts on a rim of the container body configured to contact one or more electrical contacts on the lid when the lid is coupled to the container body, the circuitry being housed in the container body and the one or more thermoelectric elements being housed in the lid, the electrical contacts facilitating control of the operation of the one or more thermoelectric elements and one or more fans by the circuitry when the lid is coupled to the container body.

(217) Clause 18. The portable cooler container of any of clauses 11-17, wherein the gap is under vacuum.

(218) Clause 19. The portable cooler container of any of clauses 11-18, further comprising means for thermally disconnecting the one or more thermoelectric elements from the chamber to inhibit heat transfer between the one or more thermoelectric elements and the chamber.

(219) Clause 20. A portable cooler container with active temperature control, comprising: a container body having a chamber configured to receive and hold one or more volumes of perishable liquid, the chamber defined by a base and an inner peripheral wall of the container body;

a lid movably coupled to the container body by one or more hinges; and a temperature control system, comprising one or more thermoelectric elements configured to actively heat or cool at least a portion of the chamber, one or more power storage elements, circuitry configured to control an operation of the one or more thermoelectric elements to heat or cool at least a portion of the chamber to a predetermined temperature or temperature range, the circuitry further configured to wirelessly communicate with a cloud-based data storage system or a remote electronic device; and an electronic display screen disposed on one or both of the container body and the lid, the display screen configured to selectively display shipping information for the portable cooler container.

(220) Clause 21. The portable cooler container of clause 20, wherein the electronic display screen is an electrophoretic display screen.

(221) Clause 22. The portable cooler container of any of clauses 20-21, further comprising a button or touch screen actuatable by a user to automatically switch sender and recipient information on the display screen to facilitate return of the portable cooler container to a sender.

(222) Clause 23. The portable cooler container of any of clauses 20-22, further comprising means for thermally disconnecting the one or more thermoelectric elements from the chamber to inhibit heat transfer between the one or more thermoelectric elements and the chamber.

(223) While certain embodiments of the inventions have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the disclosure. Indeed, the novel methods and systems described herein may be embodied in a variety of other forms. For example, though the features disclosed herein are described for medicine containers, the features are applicable to containers that are not medicine containers (e.g., portable coolers for food, etc.) and the invention is understood to extend to such other containers. Furthermore, various omissions, substitutions and changes in the systems and methods described herein may be made without departing from the spirit of the disclosure. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the disclosure. Accordingly, the scope of the present inventions is defined only by reference to the appended claims.

(224) Features, materials, characteristics, or groups described in conjunction with a particular aspect, embodiment, or example are to be understood to be applicable to any other aspect, embodiment or example described in this section or elsewhere in this specification unless incompatible therewith. All of the features disclosed in this specification (including any accompanying claims, abstract and drawings), and/or all of the steps of any method or process so disclosed, may be combined in any combination, except combinations where at least some of such features and/or steps are mutually exclusive. The protection is not restricted to the details of any foregoing embodiments. The protection extends to any novel one, or any novel combination, of the features disclosed in this specification (including any accompanying claims, abstract and drawings), or to any novel one, or any novel combination, of the steps of any method or process so disclosed.

(225) Furthermore, certain features that are described in this disclosure in the context of separate implementations can also be implemented in combination in a single implementation. Conversely, various features that are described in the context of a single implementation can also be implemented in multiple implementations separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations, one or more features from a claimed combination can, in some cases, be excised from the combination, and the combination may be claimed as a subcombination or variation of a subcombination.

(226) Moreover, while operations may be depicted in the drawings or described in the specification in a particular order, such operations need not be performed in the particular order shown or in sequential order, or that all operations be performed, to achieve desirable results. Other operations that are not depicted or described can be incorporated in the example methods and processes. For example, one or more additional operations can be performed before, after, simultaneously, or

between any of the described operations. Further, the operations may be rearranged or reordered in other implementations. Those skilled in the art will appreciate that in some embodiments, the actual steps taken in the processes illustrated and/or disclosed may differ from those shown in the figures. Depending on the embodiment, certain of the steps described above may be removed, others may be added. Furthermore, the features and attributes of the specific embodiments disclosed above may be combined in different ways to form additional embodiments, all of which fall within the scope of the present disclosure. Also, the separation of various system components in the implementations described above should not be understood as requiring such separation in all implementations, and it should be understood that the described components and systems can generally be integrated together in a single product or packaged into multiple products.

(227) For purposes of this disclosure, certain aspects, advantages, and novel features are described herein. Not necessarily all such advantages may be achieved in accordance with any particular embodiment. Thus, for example, those skilled in the art will recognize that the disclosure may be embodied or carried out in a manner that achieves one advantage or a group of advantages as taught herein without necessarily achieving other advantages as may be taught or suggested herein.

(228) Conditional language, such as “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements, and/or steps. Thus, such conditional language is not generally intended to imply that features, elements, and/or steps are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without user input or prompting, whether these features, elements, and/or steps are included or are to be performed in any particular embodiment.

(229) Conjunctive language such as the phrase “at least one of X, Y, and Z,” unless specifically stated otherwise, is otherwise understood with the context as used in general to convey that an item, term, etc. may be either X, Y, or Z. Thus, such conjunctive language is not generally intended to imply that certain embodiments require the presence of at least one of X, at least one of Y, and at least one of Z.

(230) Language of degree used herein, such as the terms “approximately,” “about,” “generally,” and “substantially” as used herein represent a value, amount, or characteristic close to the stated value, amount, or characteristic that still performs a desired function or achieves a desired result. For example, the terms “approximately,” “about,” “generally,” and “substantially” may refer to an amount that is within less than 10% of, within less than 5% of, within less than 1% of, within less than 0.1% of, and within less than 0.01% of the stated amount. As another example, in certain embodiments, the terms “generally parallel” and “substantially parallel” refer to a value, amount, or characteristic that departs from exactly parallel by less than or equal to 15 degrees, 10 degrees, 5 degrees, 3 degrees, 1 degree, or 0.1 degree.

(231) The scope of the present disclosure is not intended to be limited by the specific disclosures of preferred embodiments in this section or elsewhere in this specification, and may be defined by claims as presented in this section or elsewhere in this specification or as presented in the future. The language of the claims is to be interpreted broadly based on the language employed in the claims and not limited to the examples described in the present specification or during the prosecution of the application, which examples are to be construed as non-exclusive.

Claims

1. A portable cooler container with active temperature control for medicines, vaccines, biological samples or other medical goods, comprising: a container body having an insulated chamber configured to receive and hold one or more medicines, vaccines, biological samples or other medical goods, the container body comprising an outer peripheral wall and a bottom portion

attached to the outer peripheral wall, the insulated chamber suspended relative to the outer peripheral wall, a base spaced apart from the bottom portion to define a cavity between the base and the bottom portion; a lid removably coupleable or hingedly coupleable to the container body to access the chamber, the chamber being sealed when the lid is coupled to the container body to close the chamber; a temperature control system at least partially disposed in the container body between an outer container wall and the insulated chamber, the temperature control system comprising: one or more thermoelectric elements in thermal communication with an inner wall surface of the insulated chamber and operable to actively heat or cool at least a portion of the chamber, and circuitry configured to control an operation of the one or more thermoelectric elements to heat or cool at least a portion of the insulated chamber to a predetermined temperature or temperature range; the container body having a gap between the insulated chamber and the outer container wall that is under vacuum to provide vacuum insulation about the insulated chamber; and an electronic display screen configured to selectively display shipping information for the portable cooler container.

2. The portable cooler container of claim 1, wherein the circuitry is further configured to automatically switch sender information and recipient information on the display screen to facilitate return of the portable cooler container to a sender.

3. The portable cooler container of claim 2, further comprising one or more temperature sensors configured to sense a temperature in the insulated chamber and one or more GPS sensors configured to sense a location of the container, the one or more temperature sensors and one or more GPS sensors configured to communicate the sensed temperature and sensed location to the circuitry, the circuitry configured to store the sensed temperature and the sensed location in a memory of the container.

4. The portable cooler container of claim 3, further comprising one or more motion sensors configured to sense a motion of the container body and to communicate sensed motion data to the circuitry, the circuitry configured to store the sensed motion data in the memory of the container.

5. The portable cooler container of claim 4, wherein the circuitry stores the sensed temperature, the sensed location and the sensed motion in the memory of the container.

6. The portable cooler container of claim 1, wherein the one or more thermoelectric elements are in thermal communication with the inner wall surface of the insulated chamber through a wall of the insulated chamber and are operable to heat or cool the insulated chamber via conduction heat transfer through the wall of the insulated chamber.

7. The portable cooler container of claim 1, wherein the temperature control system further comprises a heat sink unit in thermal communication with one side of the one or more thermoelectric elements and one or more fans, the one or more fans operable to draw air through one or more air intake openings and flow air past the heat sink unit to remove heat from the heat sink unit and flow said air out through one or more exhaust openings, the circuitry configured to operate the one or more thermoelectric elements and the one or more fans to heat or cool at least a portion of the insulated chamber to the predetermined temperature or temperature range.

8. The portable cooler container of claim 7, further comprising one or more electrical connectors connectable to a power source to power one or both of the temperature control system and charge one or more batteries of the container.

9. The portable cooler container of claim 7, wherein the container body further comprises one or more electrical contacts configured to contact one or more electrical contacts of a power base when the container body is placed on the power base so that the circuitry controls the operation of the one or more thermoelectric elements and one or more fans when the container body is on the power base.

10. A portable cooler container with active temperature control for medicines, vaccines, biological samples or other medical goods, comprising: a container body having an insulated chamber configured to receive and hold one or more medicines, vaccines, biological samples or other

medical goods, the container body comprising an outer peripheral wall and a bottom portion attached to the outer peripheral wall, the insulated chamber suspended relative to the outer peripheral wall, a base of the insulated chamber spaced apart from the bottom portion to define a cavity between the base and the bottom portion; a lid hingedly coupleable to the container body to access the chamber, the chamber being sealed when the lid is coupled to the container body to close the chamber; a temperature control system at least partially disposed in the container body between an outer container wall and the insulated chamber, the temperature control system comprising: one or more thermoelectric elements in thermal communication with an inner wall surface of the insulated chamber and operable to actively heat or cool at least a portion of the chamber, and circuitry configured to control an operation of the one or more thermoelectric elements to heat or cool at least a portion of the insulated chamber to a predetermined temperature or temperature range; the container body having a gap between the insulated chamber and the outer container wall that is under vacuum to provide vacuum insulation about the insulated chamber; and an electronic display screen actuatable to display shipping address information for the portable cooler container.

11. The portable cooler container of claim 10, wherein the circuitry is further configured to automatically switch sender information and recipient information on the display screen to facilitate return of the portable cooler container to a sender.

12. The portable cooler container of claim 11, further comprising one or more temperature sensors configured to sense a temperature in the insulated chamber and one or more GPS sensors configured to sense a location of the container, the one or more temperature sensors and one or more GPS sensors configured to communicate the sensed temperature and the sensed location to the circuitry, the circuitry configured to store the sensed temperature and the sensed location in a memory of the container.

13. The portable cooler container of claim 12, further comprising one or more motion sensors configured to sense a motion of the container body and to communicate sensed motion data to the circuitry, the circuitry configured to store the sensed motion data in the memory of the container.

14. The portable cooler container of claim 13, wherein the circuitry stores the sensed temperature, the sensed location and the sensed motion in the memory of the container.

15. The portable cooler container of claim 10, wherein the temperature control system further comprises a heat sink unit in thermal communication with one side of the one or more thermoelectric elements and one or more fans, the one or more fans operable to draw air through one or more air intake openings and flow air past the heat sink unit to remove heat from the heat sink unit and flow said air out through one or more exhaust openings, the circuitry configured to operate the one or more thermoelectric elements and the one or more fans to heat or cool at least a portion of the insulated chamber to the predetermined temperature or temperature range.

16. The portable cooler container of claim 15, further comprising one or more electrical connectors connectable to a power source to power one or both of the temperature control system and charge one or more batteries of the container.

17. The portable cooler container of claim 10, wherein the one or more thermoelectric elements are in thermal communication with the inner wall surface of the insulated chamber through a wall of the insulated chamber and are operable to heat or cool the insulated chamber via conduction heat transfer through the wall of the insulated chamber.

18. A portable cooler container with active temperature control for medicines, vaccines, biological samples or other medical goods, comprising: a container body having a vacuum insulated chamber configured to receive and hold one or more medicines, vaccines, biological samples or other medical goods, the container body comprising an outer peripheral wall and a bottom portion attached to the outer peripheral wall, the insulated chamber suspended relative to the outer peripheral wall, a base spaced apart from the bottom portion to define a cavity between the base and the bottom portion; a lid removably coupleable or hingedly coupleable to the container body to access the chamber, the chamber being sealed when the lid is coupled to the container body to close

the chamber; a temperature control system at least partially disposed in the container body between an outer container wall and the vacuum insulated chamber, the temperature control system comprising: one or more thermoelectric elements in thermal communication with an inner wall surface of the vacuum insulated chamber and operable to actively heat or cool at least a portion of the chamber, and circuitry configured to control an operation of the one or more thermoelectric elements to heat or cool at least a portion of the vacuum insulated chamber to a predetermined temperature or temperature range; the container body having a gap between the insulated chamber and the outer container wall that is under vacuum to provide vacuum insulation about the insulated chamber; and an electronic display screen configured to selectively display information.
